

Chapter 16

Acid–Base Equilibria

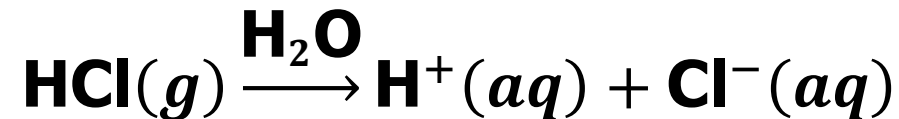
- ▶ Acid–Base Equilibria
- ▶ The Autoionization of Water
- ▶ The pH Scale
- ▶ Strong Acids and Bases
- ▶ Weak Acids
- ▶ Weak Bases
- ▶ Acid–Base Properties of Salt Solutions
- ▶ Acid–Base Behavior and Chemical Structure

Acid–Base Equilibria

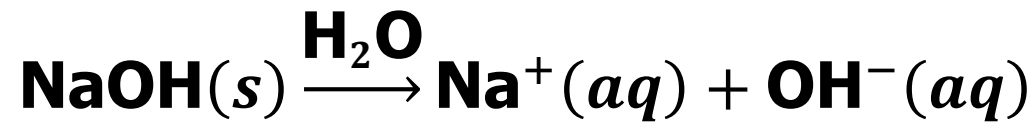
Q. How do you define Acids? Bases?

Arrhenius Acids and Bases (1880s)

- ✓ An acid is a substance that, when dissolved in water, increases the concentration of hydrogen ions.



- ✓ A base is a substance that, when dissolved in water, increases the concentration of hydroxide ions.

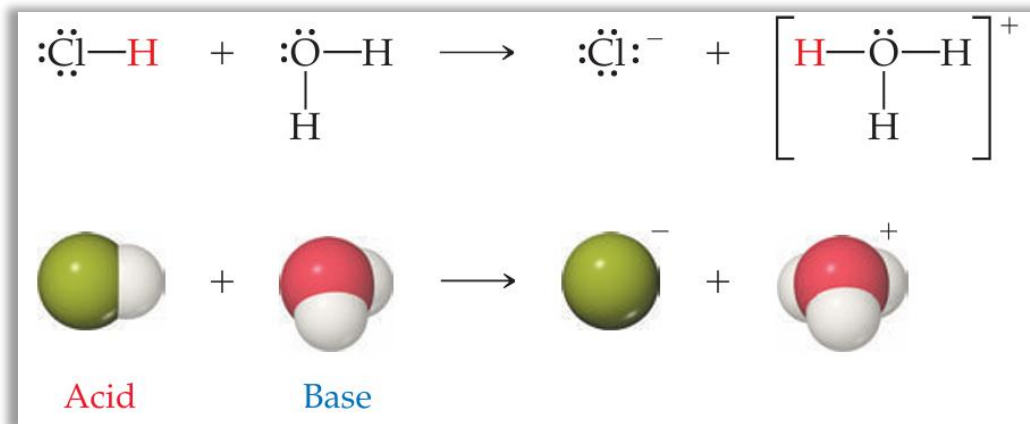
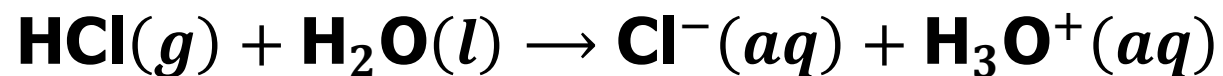


Acid–Base Equilibria

Q. How do you define Acids? Bases?

Brønsted–Lowry Acids and Bases (1923)

- Acid-base reactions involve the transfer of H^+ ions from one substance to another.
 - ✓ An acid is a substance (molecule or ion) that donates a proton to another substance.
 - ✓ A base is a substance that accepts a proton



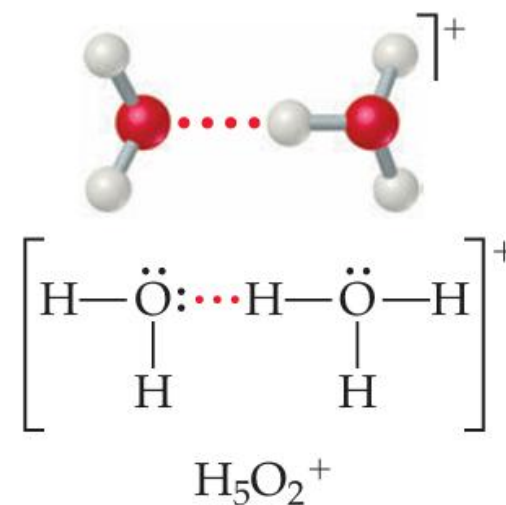
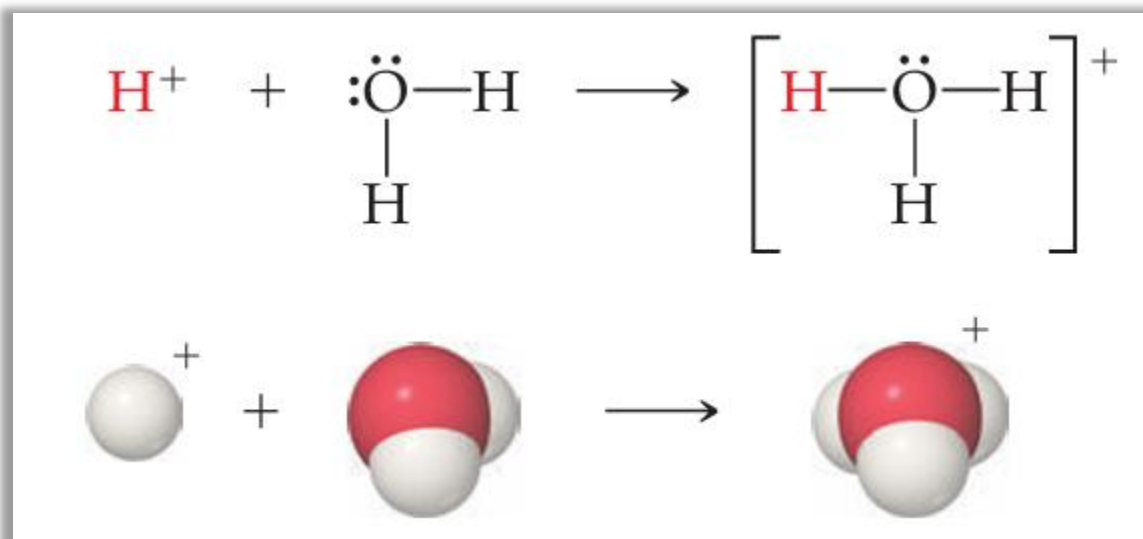
Acid–Base Equilibria

Q. How do you define Acids? Bases?

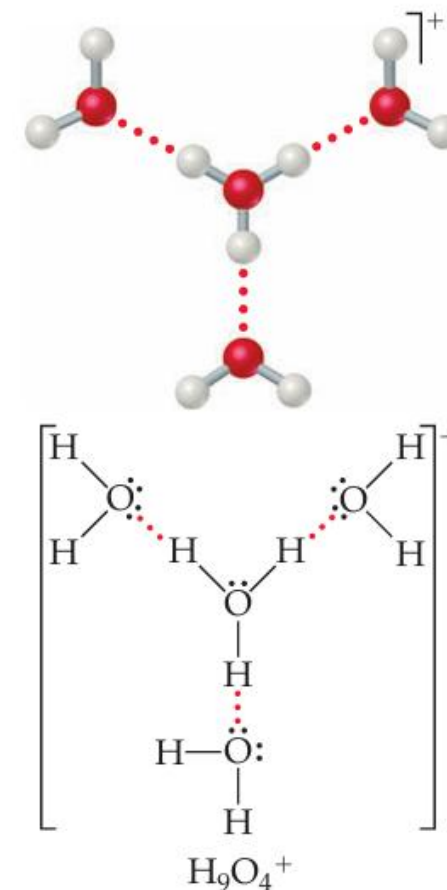
Brønsted–Lowry Acids and Bases (1923)

Hydronium ion ($\text{H}_3\text{O}^+(\text{aq})$):

- Formed by the interaction of a proton with water.



Hydrogen bond formation with water

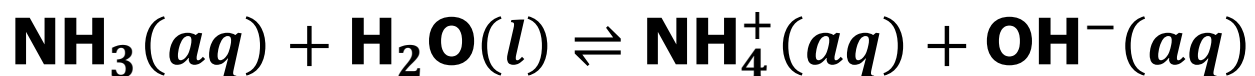
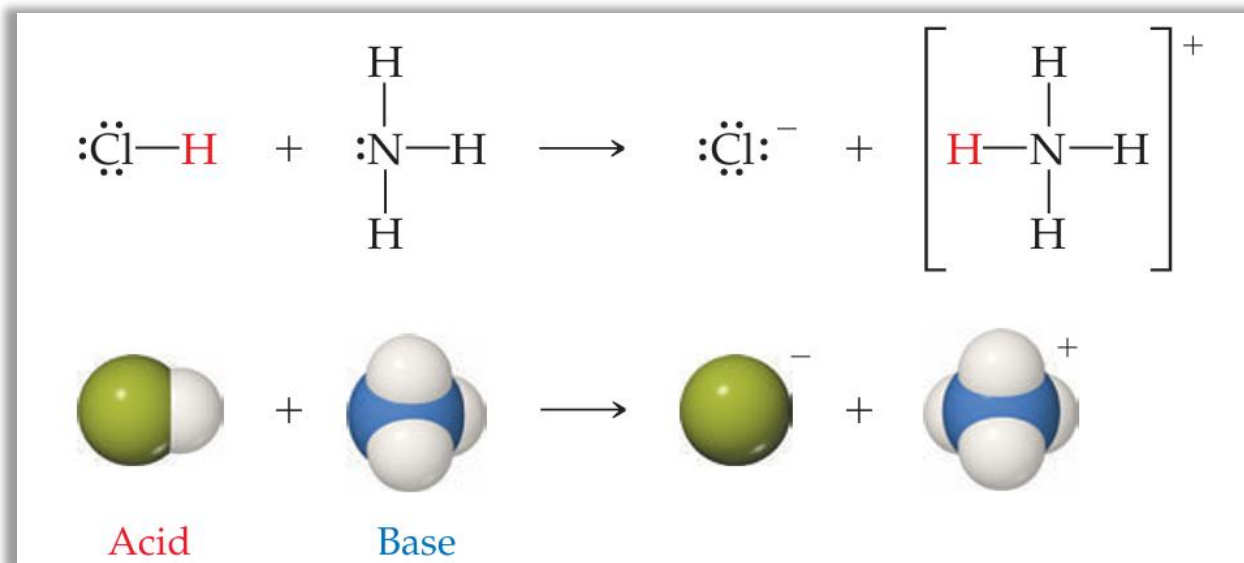
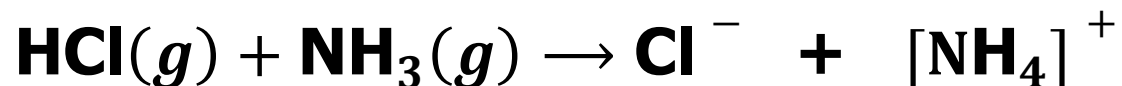


▲ Figure 16.1 Ball-and-stick models and Lewis structures for two hydrated hydronium ions.

Acid–Base Equilibria

Q. How do you define Acids? Bases?

Brønsted–Lowry Acids and Bases (1923)



▲ **Figure 16.2** Fog of NH₄Cl(s) caused by the reaction of HCl(g) and NH₃(g).

Acid–Base Equilibria

Q. How do you define Acids? Bases?

Water: Amphiprotic property

- The transfer of a proton always involves both an acid (donor) and a base (acceptor).

$\text{HCl}(g) + \text{H}_2\text{O}(l) \rightarrow \text{Cl}^-(aq) + \text{H}_3\text{O}^+(aq)$; Water acts as a base

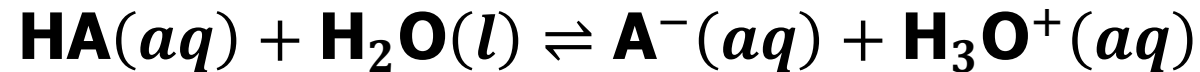
$\text{NH}_3(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{NH}_4^+(aq) + \text{OH}^-(aq)$; Water acts as an acid

- A substance capable of acting as either an acid or a base is called amphiprotic.

Acid–Base Equilibria

Conjugate Acids and Bases

Consider the acid–base equilibrium:



Forward Reaction:

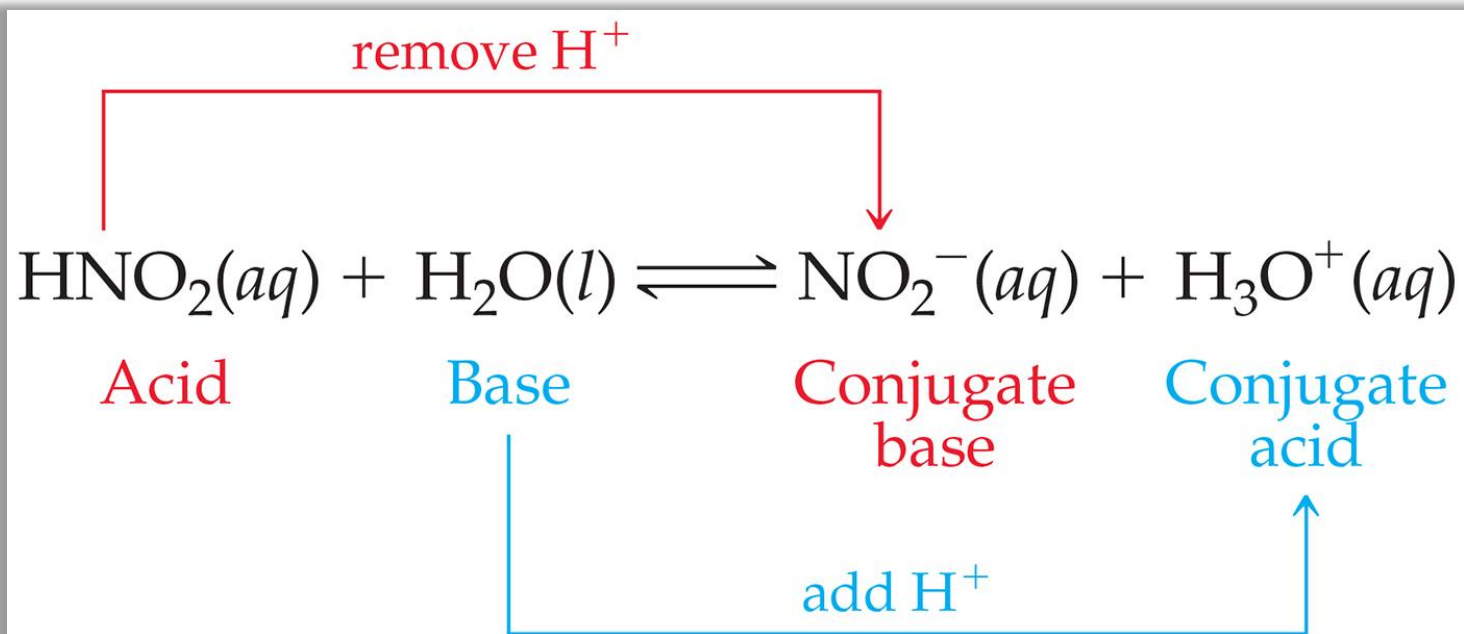
- ✓ HA: Donates a proton → Brønsted–Lowry Acid
- ✓ H₂O: Accepts a proton → Brønsted–Lowry Base

Reverse Reaction:

- ✓ H₃O⁺: Donates a proton → Acid
 - ✓ A[−]: Accepts a proton → Base
- An acid and a base such as HA and A[−] that differ only in the presence or absence of a proton are called a **conjugate acid–base pair**.

Acid–Base Equilibria

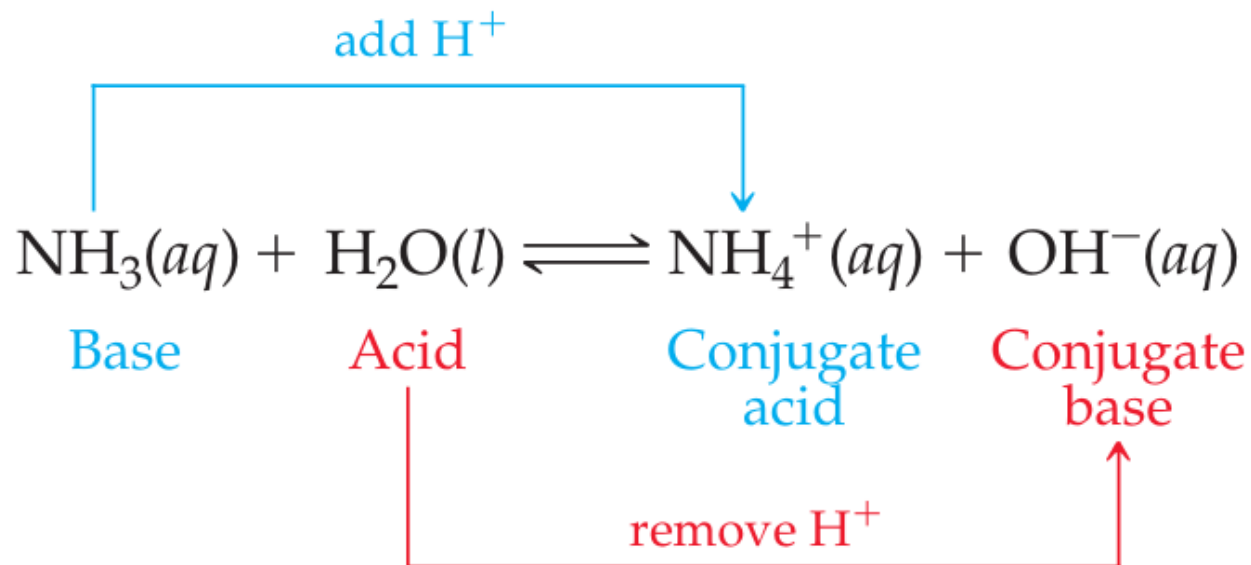
Conjugate Acids and Bases



- Any acid–base (proton-transfer) reaction features **two sets** of conjugate acid–base pairs.
 - ✓ Every acid has a conjugate base (formed by removing a proton from the acid).
 - ✓ Every base has a conjugate acid (formed by adding a proton to the base).

Acid–Base Equilibria

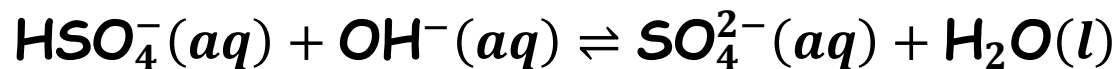
Conjugate Acids and Bases



Q. What is the conjugate base of HClO_4 , H_2S , PH_4^+ , HCO_3^- ?

Q. What is the conjugate acid of CN^- , SO_4^{2-} , H_2O , HCO_3^- ?

Q. Identify the acid, base, conjugate acid, and conjugate base in the following acid-base equilibrium.



Acid–Base Equilibria

Relative Strengths of Acids and Bases

	ACID		BASE	
Strong acids	HCl	↑ Base strength increasing ↓ ↑ Acid strength increasing ↓	Cl ⁻	Negligible basicity
	H ₂ SO ₄		HSO ₄ ⁻	
	HNO ₃		NO ₃ ⁻	
	H ₃ O ⁺ (aq)		H ₂ O	
	HSO ₄ ⁻		SO ₄ ²⁻	
Weak acids	H ₃ PO ₄		H ₂ PO ₄ ⁻	Weak bases
	HF		F ⁻	
	CH ₃ COOH		CH ₃ COO ⁻	
	H ₂ CO ₃		HCO ₃ ⁻	
	H ₂ S		HS ⁻	
	H ₂ PO ₄ ⁻		HPO ₄ ²⁻	
	NH ₄ ⁺	NH ₃		
	HCO ₃ ⁻	CO ₃ ²⁻		
	HPO ₄ ²⁻	PO ₄ ³⁻		
	Negligible acidity	H ₂ O	OH ⁻	
OH ⁻		O ²⁻		
H ₂		H ⁻		
CH ₄		CH ₃ ⁻		

Relative strengths of select conjugate acid–base pairs.

- Acids above the line with H₂O as a base are strong acids; their conjugate bases do not act as acids in water.
- Bases below the line with H₂O as an acid are strong bases; their conjugate acids do not act as acids in water.
 - ✓ The stronger an acid, the weaker its conjugate base
 - ✓ The stronger a base, the weaker its conjugate acid.

Acid–Base Equilibria

Relative Strengths of Acids and Bases

	ACID		BASE	
Strong acids	HCl		Cl ⁻	Negligible basicity
	H ₂ SO ₄		HSO ₄ ⁻	
	HNO ₃		NO ₃ ⁻	
	H ₃ O ⁺ (aq)		H ₂ O	
	HSO ₄ ⁻		SO ₄ ²⁻	
	H ₃ PO ₄		H ₂ PO ₄ ⁻	
	HF		F ⁻	
Weak acids	CH ₃ COOH		CH ₃ COO ⁻	Weak bases
	H ₂ CO ₃		HCO ₃ ⁻	
	H ₂ S		HS ⁻	
	H ₂ PO ₄ ⁻		HPO ₄ ²⁻	
	NH ₄ ⁺		NH ₃	
	HCO ₃ ⁻		CO ₃ ²⁻	
	HPO ₄ ²⁻		PO ₄ ³⁻	
	H ₂ O		OH ⁻	
Negligible acidity	OH ⁻		O ²⁻	Strong bases
	H ₂		H ⁻	
	CH ₄		CH ₃ ⁻	

↑ Acid strength increasing
 ↓ Base strength increasing

Relative strengths of select conjugate acid–base pairs.

Strong Acids

- Complete dissociation.
- Conjugate Base: Negligible basicity (negligible tendency to accept protons).

Weak Acids

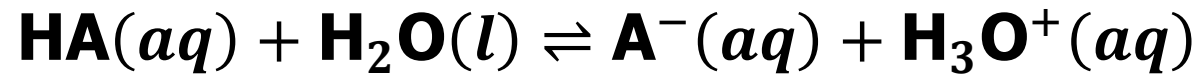
- Partial dissociation (exists as a mixture of the undissociated acid and its conjugate base).
- Conjugate Base: Weak base.

Negligible Acidity

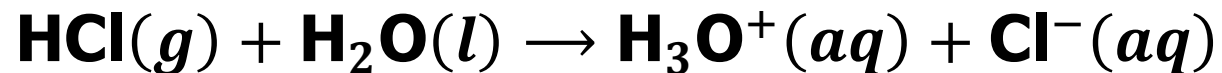
- Contains hydrogen but demonstrates no acidic behavior in water.
- Conjugate Base: Strong base.

Acid–Base Equilibria

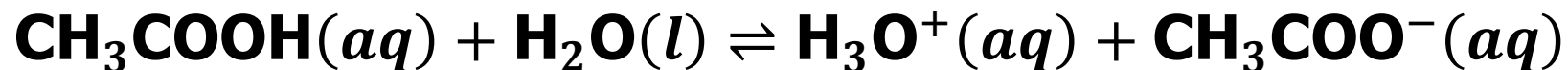
Acid-Base equilibrium: leveling effect



- Stronger acids react with water to produce $\text{H}_3\text{O}^+(aq)$ ions.
- Stronger bases react with water to produce $\text{OH}^-(aq)$ ions
- In acid–base reaction, equilibrium favors transfer of the proton from the stronger acid to the stronger base to form the weaker acid and the weaker base.



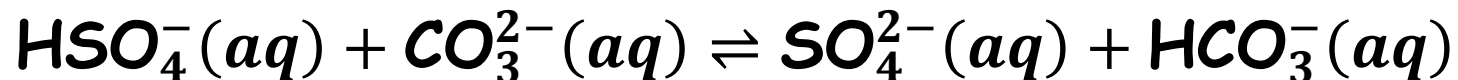
- H_2O is a much stronger base than Cl^- , so the equilibrium lies far to the right ($K \gg 1$).



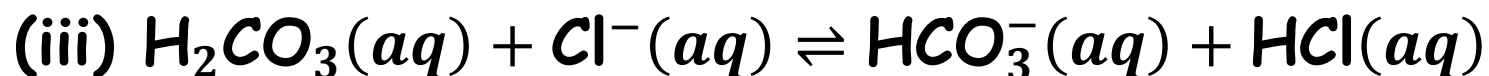
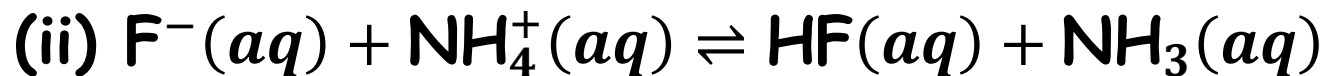
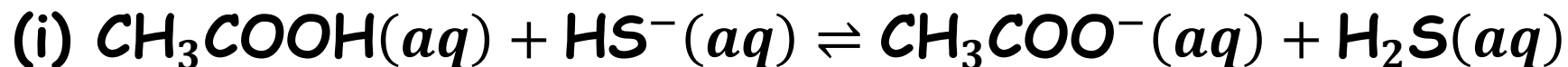
- CH_3COO^- ion is a stronger base than H_2O , so the equilibrium favors the left side ($K < 1$).

Acid–Base Equilibria

- Q. For the following proton-transfer reaction, predict whether the equilibrium lies to the left ($K_c < 1$) or to the right ($K_c > 1$):

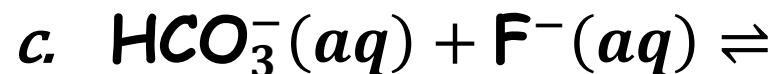
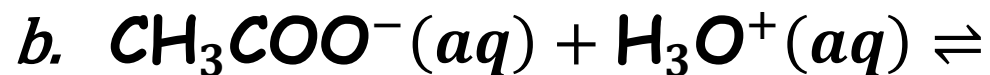
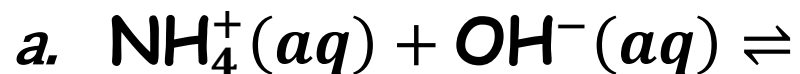


- Q. Place the following equilibria in order from smallest to largest value of K_c :

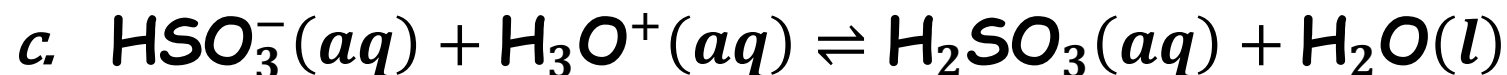
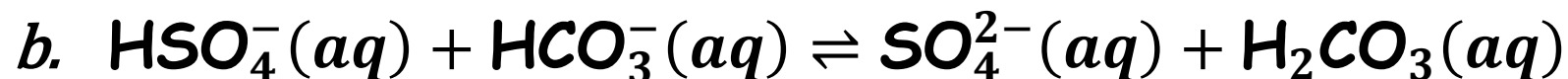
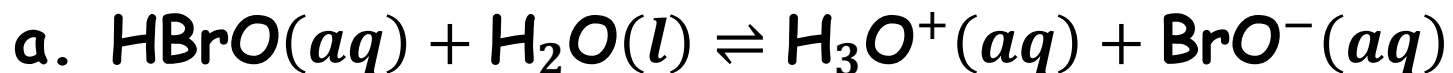


Acid–Base Equilibria

Q. Predict the products of the following acid-base reactions, and predict whether the equilibrium lies to the left or to the right of the reaction arrow:



Q. Identify the acid, base, conjugate acid, and conjugate base in each of the following equilibria:



The Autoionization of Water

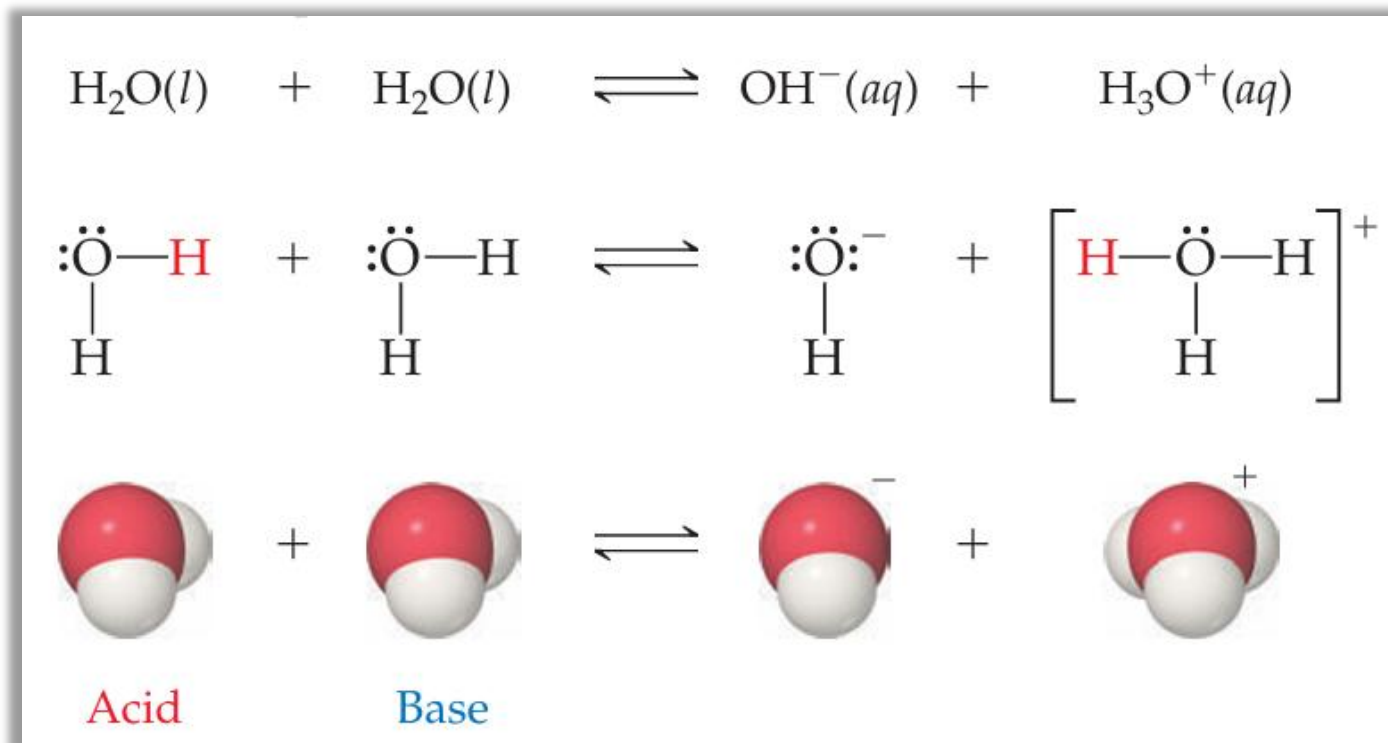
Q. What makes water act as an acid and a base?

$\text{HCl}(g) + \text{H}_2\text{O}(l) \rightarrow \text{Cl}^-(aq) + \text{H}_3\text{O}^+(aq)$; Water acts as a base

$\text{NH}_3(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{NH}_4^+(aq) + \text{OH}^-(aq)$; Water acts as an acid

- One water molecule can donate a proton to another water molecule
- We call this process the **autoionization** of water.

Q. Why is water a poor conductor of electricity?



The Autoionization of Water

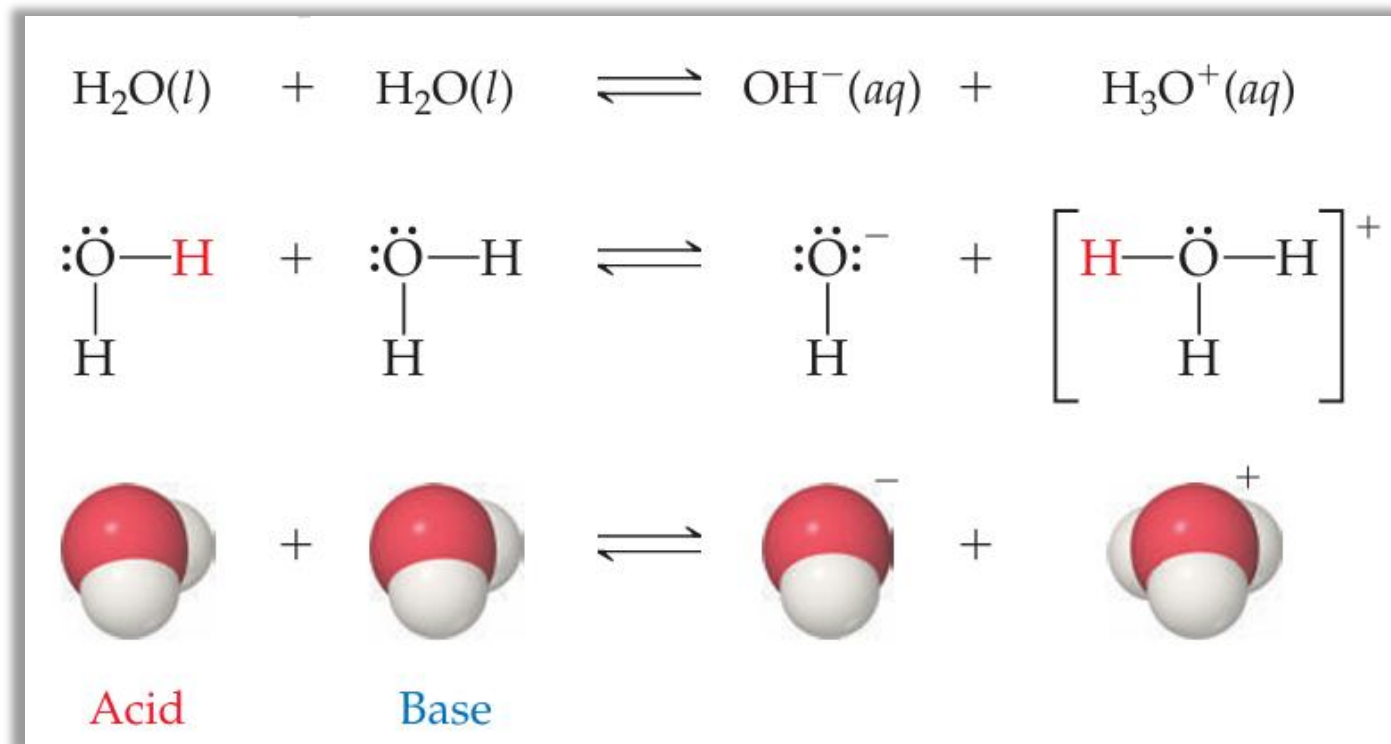
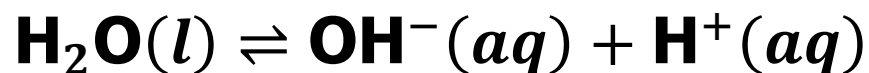
The Ion Product of Water

- The equilibrium expression for the autoionization of water is:

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-]$$

$$K_w = [\text{H}^+][\text{OH}^-]$$

$$K_w = 1.0 \times 10^{-14} \text{ at } 25^\circ\text{C}$$



Autoionization of water

The Autoionization of Water

The Ion Product of Water

$$K_w = [\text{H}^+][\text{OH}^-]$$

$$K_w = 1.0 \times 10^{-14} \text{ at } 25^\circ\text{C}$$

- $[\text{H}^+] = [\text{OH}^-]$: Neutral solution
- $[\text{H}^+] > [\text{OH}^-]$: Acidic solution
- $[\text{H}^+] < [\text{OH}^-]$: Basic solution



Acidic solution

$$[\text{H}^+] > [\text{OH}^-]$$

$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$



Neutral solution

$$[\text{H}^+] = [\text{OH}^-]$$

$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$



Basic solution

$$[\text{H}^+] < [\text{OH}^-]$$

$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$

Relative concentrations of H^+ and OH^- in aqueous solutions at 25°C

The Autoionization of Water

Q. Calculate the values of $[H^+]$ and $[OH^-]$ in a neutral aqueous solution at 25 °C.

$$K_w = [H^+][OH^-]$$

$$K_w = 1.0 \times 10^{-14} \text{ at } 25 \text{ }^\circ\text{C}$$

$$[H^+][OH^-] = 1.0 \times 10^{-14}$$

$$(x)(x) = 1.0 \times 10^{-14}$$

$$x^2 = 1.0 \times 10^{-14}$$

$$x = 1.0 \times 10^{-7} \text{ M} = [H^+] = [OH^-]$$

Q. In a certain acidic solution at 25 °C, $[H^+]$ is 100 times greater than $[OH^-]$. What is the value for $[OH^-]$ for the solution?

The Autoionization of Water

- Q. Calculate the concentration of H^+ (aq) in the following solutions at 25 °C.
- A solution in which $[OH^-]$ is 0.010 M.
 - A solution in which $[OH^-]$ is $1.8 \times 10^{-9}M$.
- Q. A solution has $[OH^-] = 4.0 \times 10^{-8}$ at 25 °C. What is the value of $[H^+]$ for the solution?
- Q. At our body temperature of 37 °C, $K_w = 2.1 \times 10^{-14}$. What is $[H^+]$ and $[OH^-]$?
- Q. Calculate $[OH^-]$ for each of the following solutions, and indicate whether the solution is acidic, basic, or neutral:
- $[H^+] = 0.00010 M$
 - $[H^+] = 7.3 \times 10^{-14} M$
 - A solution in which $[OH^-]$ is 100 times greater than $[H^+]$.

The pH Scale

Q. What is the use of pH?

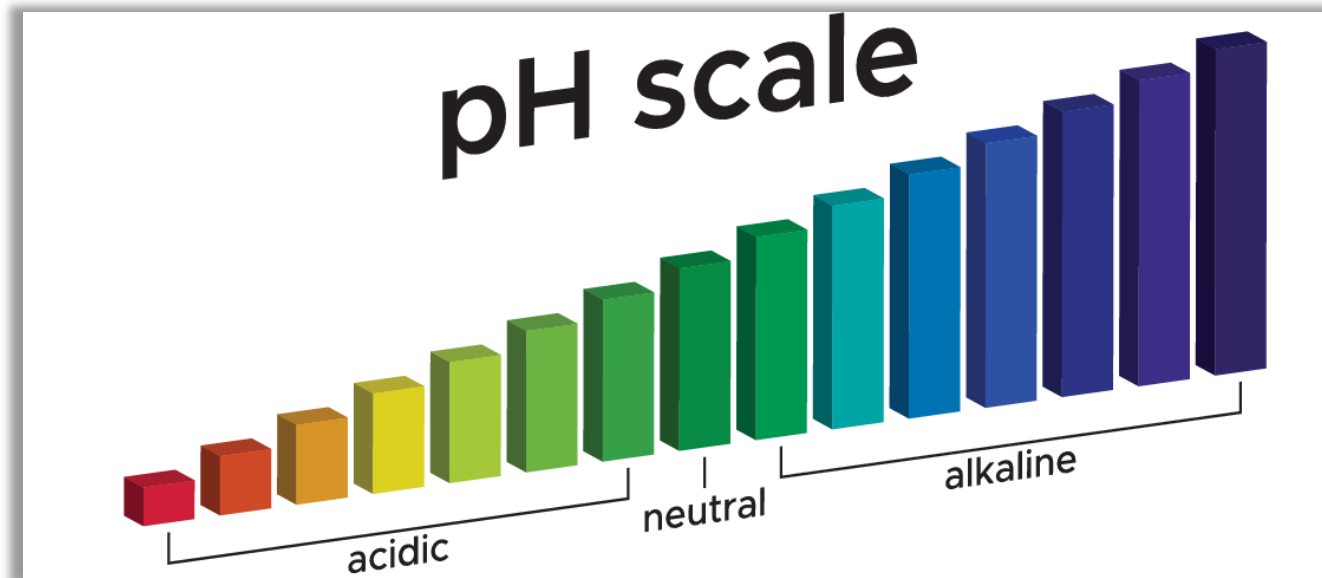
- The molar concentration of $[H^+]$ in an aqueous solution is usually very small.
- For convenience, $[H^+]$ is expressed in terms of pH (the negative logarithm in base 10 of $[H^+]$).

$$\text{pH} = -\log[H^+] = -\log[H_3O^+]$$

Example:

✓ For a neutral aqueous solution at 25 °C, $[H^+] = 1.0 \times 10^{-7} \text{ M}$.

$$\text{pH} = -\log(1.0 \times 10^{-7}) = -(-7.00) = 7.00$$



The pH Scale

Q. What is the use of pH?

$$\text{pH} = -\log[\text{H}^+]$$

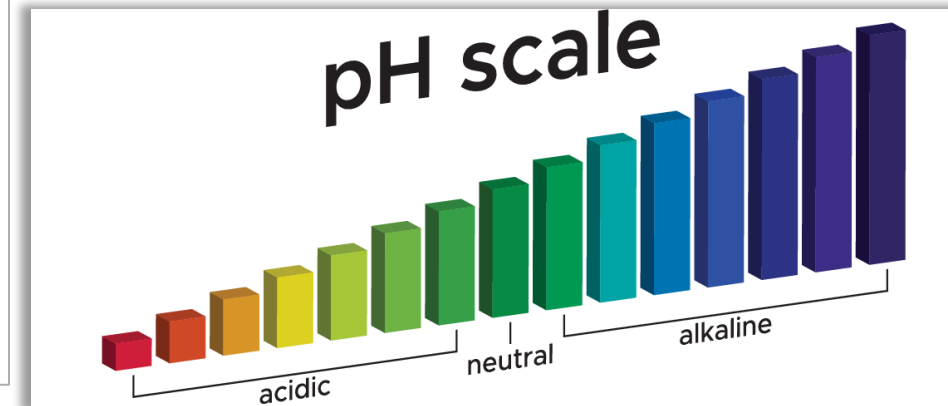
Q. As a solution becomes more acidic (so that $[\text{H}^+]$ increases), what will happen to the pH?

Q. As a solution becomes more basic ($[\text{OH}^-] > 7 \times 10^{-7} \text{ M}$), what will happen to the pH?

Suppose $[\text{OH}^-] = 2.0 \times 10^{-3} \text{ M}$:

$$[\text{H}^+] = \frac{K_w}{[\text{OH}^-]} = \frac{1.0 \times 10^{-14}}{2.0 \times 10^{-3}} = 5.0 \times 10^{-12} \text{ M}$$

$$\text{pH} = -\log(5.0 \times 10^{-12}) = \mathbf{11.30}$$



- At 25 °C the pH of an acidic solution is less than 7.00.
- At 25 °C the pH of a basic solution is greater than 7.00

The pH Scale

TABLE 16.1 Relationships between $[H^+]$, $[OH^-]$, and pH at 25 °C for Acidic, Neutral, and Basic Aqueous Solutions

	Acidic	Neutral	Basic
pH	<7.00	7.00	>7.00
$[H^+](M)$	$>1.0 \times 10^{-7}$	1.0×10^{-7}	$<1.0 \times 10^{-7}$
$[OH^-](M)$	$<1.0 \times 10^{-7}$	1.0×10^{-7}	$>1.0 \times 10^{-7}$

Q. Calculate the pH in the following solutions at 25 °C.

- A solution in which $[OH^-]$ is 0.010 M. (*Ans. 12.00*)
- A solution in which $[OH^-]$ is $1.8 \times 10^{-9}M$.
- A solution in which $[OH^-]$ is $6.7 \times 10^{-3}M$.

The pH Scale

pOH and Other “p” Scales

$$\text{pH} = -\log[\text{H}^+]$$

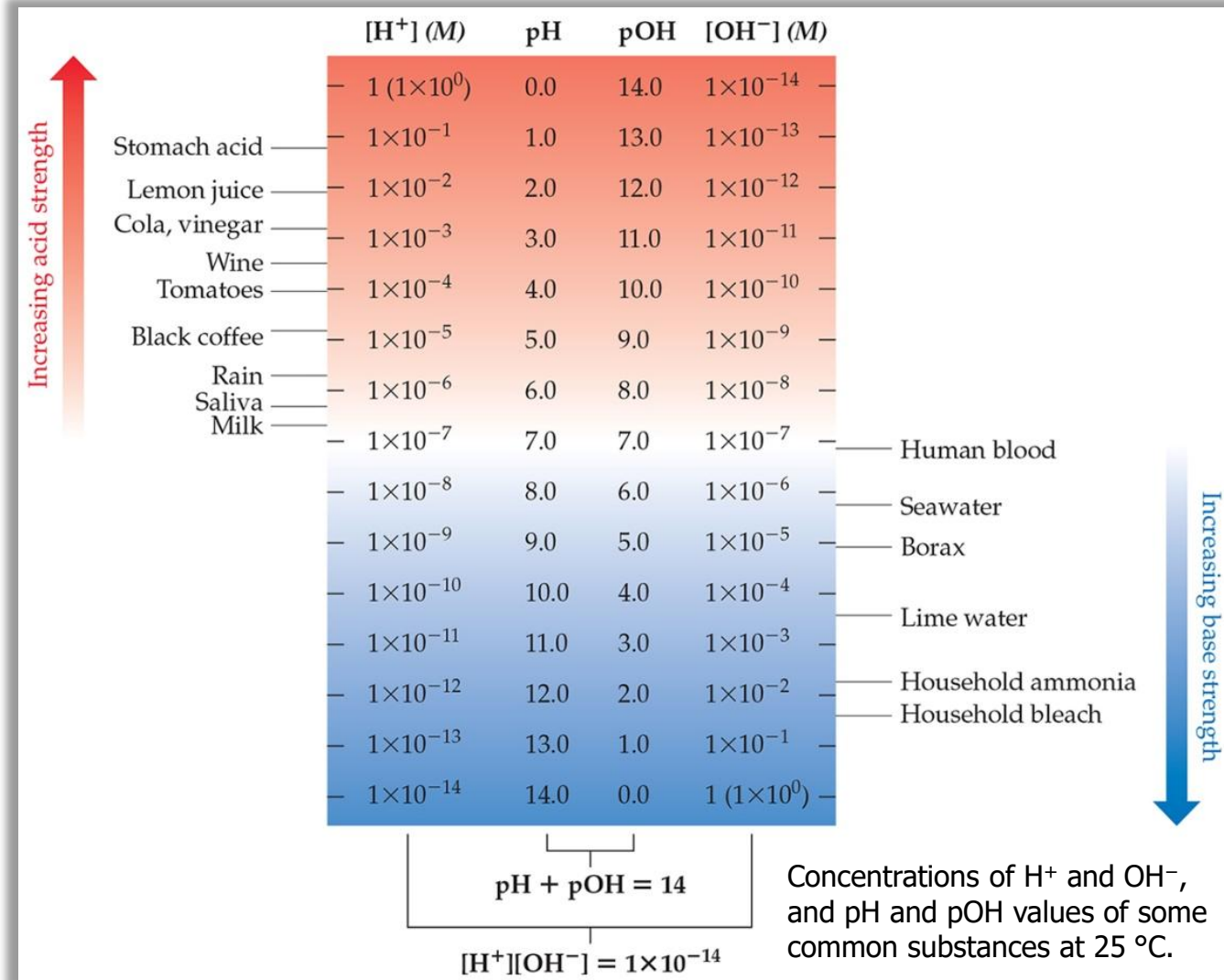
$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{p}K_w = -\log K_w$$

$$K_w = [\text{H}^+][\text{OH}^-]$$

$$-\log[\text{H}^+] + (-\log[\text{OH}^-]) = -\log K_w$$

$$\text{pH} + \text{pOH} = 14.00 \quad (\text{at } 25^\circ\text{C})$$



The pH Scale

How Do We Measure pH?

1. pH meter:

- pH meters are used for accurate measurement of pH.
- Electrodes indicate small changes in voltage to detect pH.



A digital pH meter.

The pH Scale

How Do We Measure pH?

2. Indicators:

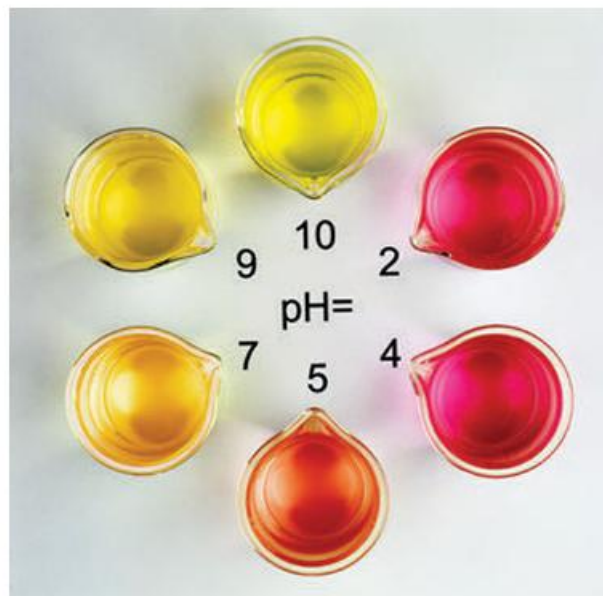
- They give less accurate but quick measurements.
- An indicator is a compound that has one color in its acid form and another color in its basic form.

pH ranges for common acid–base indicators

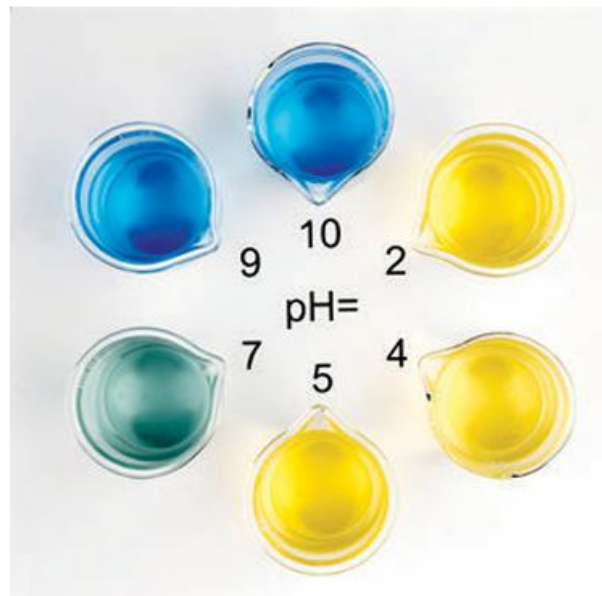
	pH range for color change										
	0	2	4	6	8	10	12	14			
Methyl violet	Yellow		Violet								
Thymol blue	Red		Yellow			Yellow		Blue			
Methyl orange			Red		Yellow						
Methyl red				Red		Yellow					
Bromthymol blue					Yellow		Blue				
Phenolphthalein						Colorless		Pink			
Alizarin yellow R								Yellow		Red	

The pH Scale

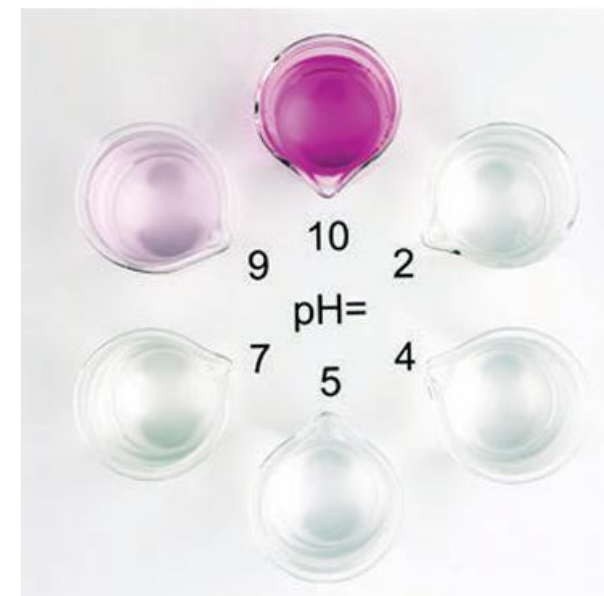
How Do We Measure pH?



Methyl red



Bromthymol blue



Phenolphthalein

Solutions containing three common acid–base indicators at various pH values

Q. Which of these indicators is best suited to distinguish between a solution that is slightly acidic and one that is slightly basic?

Strong Acids and Bases

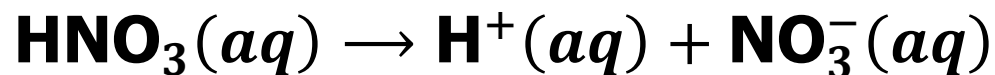
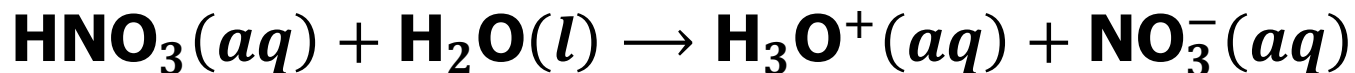
Q. What determines the strength of an acid or a base?

Strong Acids

- The seven strong acids are:
 - ✓ HCl, HBr, HI, HNO₃, H₂SO₄, HClO₃, HClO₄

Q. Identify which of these acids are monoprotic or diprotic.

- These are strong electrolytes and exist totally as ions in aqueous solution.



Q. How do we determine the pH of these acids?

Strong Acids and Bases

Q. How do we determine the pH of these acids?

Strong Acids

- In aqueous solutions of strong acids, the acid is the dominant source of **hydronium ions**.
- **Monoprotic Acids:** Calculating $[H^+]$ is straightforward ($[H^+] = [\text{acid}]_{\text{initial}}$)

Example: In a 0.20 M solution of $HNO_3(aq)$: $[H^+] = [NO_3^-] = 0.20\text{ M}$

Q. What is the pH of a 0.040 M solution of $HClO_4$?

- **Polyprotic/Diprotic Acids:** Acids like H_2SO_4 exhibit more complex behavior due to sequential dissociation steps.

Q. Show the dissociation of H_2SO_4 and propose how its pH can be determined?

Strong Acids and Bases

Strong Bases

- Ionic hydroxides of alkali metals (e.g., NaOH, KOH) and heavier alkaline earth metals (e.g., $\text{Sr}(\text{OH})_2$)
- Completely dissociate into ions in aqueous solution.
- **Example:** A 0.30 M NaOH solution consists of: 0.30 M Na^+ and 0.30 M OH^- (Essentially no undissociated NaOH present)
- Alkali metal hydroxides and alkaline earth hydroxides are **strong electrolytes**.
 - ✓ LiOH, RbOH, and CsOH (not commonly encountered in the lab).
 - ✓ $\text{Ca}(\text{OH})_2$, $\text{Sr}(\text{OH})_2$, and $\text{Ba}(\text{OH})_2$.
- Alkaline earth hydroxides have limited solubility.

Strong Acids and Bases

Strong Bases

- Basic solutions can also be prepared via metal oxides.
- Ionic metal oxides (Na_2O , CaO) are frequently used in industry.
- Reaction mechanism involves exothermic reaction of O^{2-} with water to form OH^- .



Q. What is the pH of 0.028 M solution of NaOH?

✓ $\text{pOH} = -\log(0.028) = 1.55$

✓ $\text{pH} = 14.00 - \text{pOH} = 12.45$

Q. What is the pH of 0.0011 M solution of $\text{Ca}(\text{OH})_2$?

Q. Order the following three solutions from smallest to largest pH:

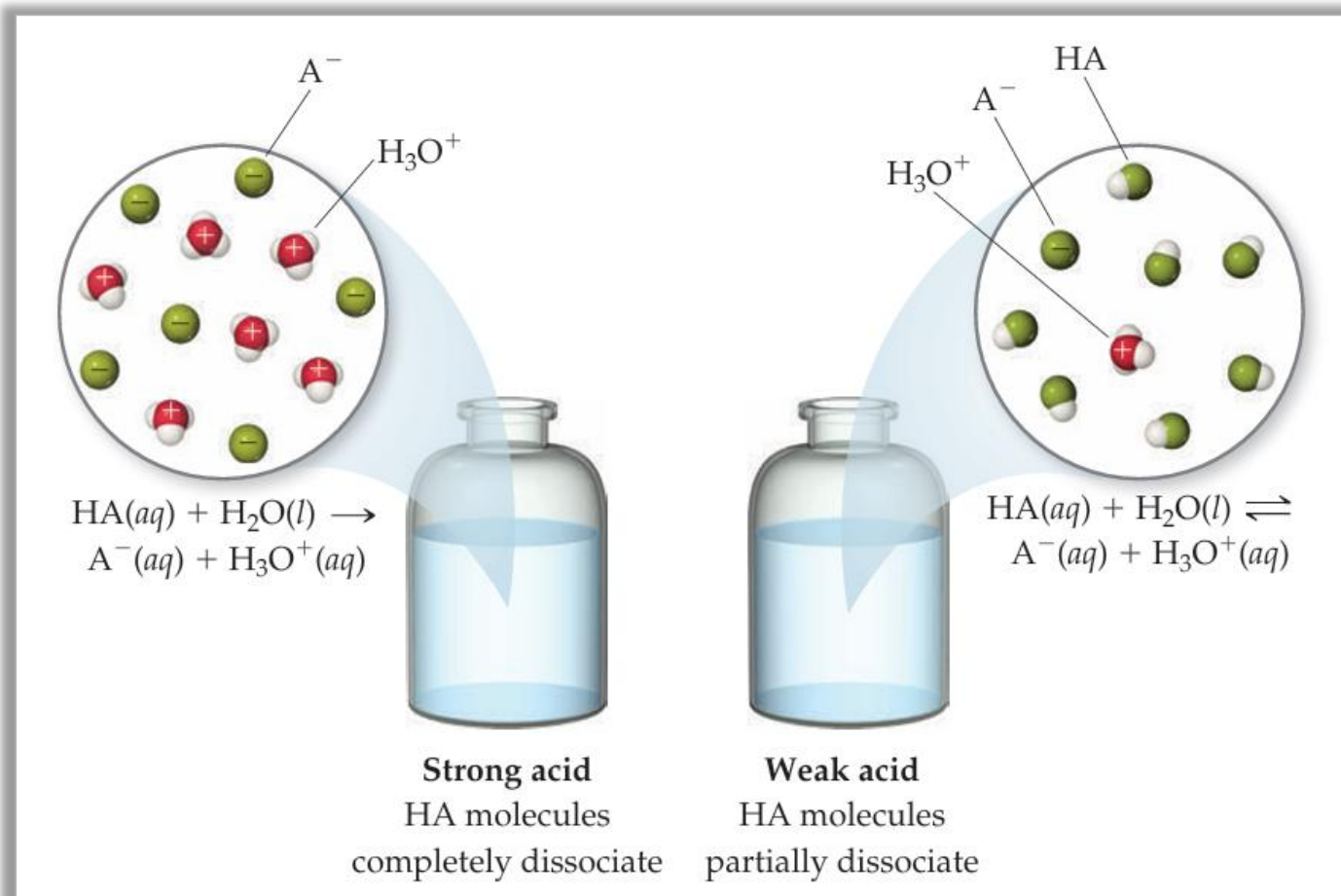
(i) 0.030 M $\text{Ba}(\text{OH})_2$

(ii) 0.040 M KOH

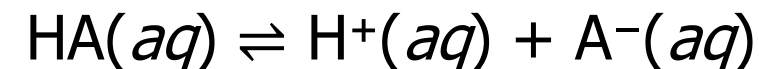
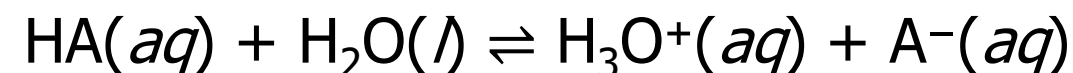
(iii) pure water.

Weak Acids

Q. How are weak acids different from strong acids?



- Weak acids are only partially ionized in aqueous solution



$$K_a = \frac{[H_3O^+][A^-]}{[HA]} \quad \text{or}$$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

K_a : acid-dissociation constant

Species present in solutions of a strong acid and a weak acid.

Weak Acids

- The larger the value of K_a , the stronger the acid.

TABLE 16.2 Some Weak Acids in Water at 25 °C

Acid	Structural Formula*	Conjugate Base	K_a
Chlorous (HClO ₂)	$\text{H}-\text{O}-\text{Cl}-\text{O}$	ClO_2^-	1.0×10^{-2}
Hydrofluoric (HF)	$\text{H}-\text{F}$	F^-	6.8×10^{-4}
Nitrous (HNO ₂)	$\text{H}-\text{O}-\text{N}=\text{O}$	NO_2^-	4.5×10^{-4}
Benzoic (C ₆ H ₅ COOH)	$\text{H}-\text{O}-\text{C}(=\text{O})-\text{C}_6\text{H}_5$	$\text{C}_6\text{H}_5\text{COO}^-$	6.3×10^{-5}
Acetic (CH ₃ COOH)	$\text{H}-\text{O}-\text{C}(=\text{O})-\text{CH}_3$	CH_3COO^-	1.8×10^{-5}
Hypochlorous (HOCl)	$\text{H}-\text{O}-\text{Cl}$	OCl^-	3.0×10^{-8}
Hydrocyanic (HCN)	$\text{H}-\text{C}\equiv\text{N}$	CN^-	4.9×10^{-10}
Phenol (HOC ₆ H ₅)	$\text{H}-\text{O}-\text{C}_6\text{H}_5$	$\text{C}_6\text{H}_5\text{O}^-$	1.3×10^{-10}

*The proton that ionizes is shown in red.

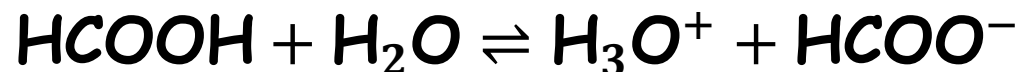
Extended K_a value: Appendix D in the textbook

- Q. Identify the strongest acid from this table.
- Q. Identify the weakest acid from this table.
- Q. Order all acids by acidity level.
- Q. Identify the strongest conjugate base from this table.

Weak Acids

Calculating K_a from pH

- The pH of a 0.10 M solution of formic acid, HCOOH , at 25 °C is 2.38. Calculate K_a for formic acid at this temperature.



1

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{HCOO}^-]}{[\text{HCOOH}]}$$

3

	$[\text{HCOOH}], \text{M}$	$[\text{H}_3\text{O}^+], \text{M}$	$[\text{HCOO}^-], \text{M}$
Initial	0.10	0	0
Change	-4.2×10^{-3}	$+4.2 \times 10^{-3}$	$+4.2 \times 10^{-3}$
Equilibrium	$0.10 - 4.2 \times 10^{-3} = 0.10$	4.2×10^{-3}	4.2×10^{-3}

2

$$\checkmark \text{ pH} = -\log[\text{H}^+] = 2.38$$

$$\checkmark \log[\text{H}^+] = -2.38$$

$$\checkmark [\text{H}^+] = 10^{-2.38} = 4.2 \times 10^{-3} \text{ M}$$

4

$$K_a = \frac{(4.2 \times 10^{-3})(4.2 \times 10^{-3})}{0.10} = 1.8 \times 10^{-4}$$

Weak Acids

Percent Ionization

- Apart from K_a , percent ionization can be used to measure acid strength.
- The stronger the acid, the greater the percent ionization.

$$\text{Percent ionization} = \frac{\text{Concentration of ionized HA}}{\text{Original concentration of HA}} \times 100\%$$

- Assuming the autoionization of H_2O is negligible:

$$\text{Percent ionization} = \frac{[\text{H}^+]_{\text{equilibrium}}}{[\text{HA}]_{\text{initial}}} \times 100\%$$

For example, a 0.035 M solution of HNO_2 contains $3.7 \times 10^{-3} \text{ M H}^+(\text{aq})$:

$$\text{Percent ionization} = \frac{[\text{H}^+]_{\text{equilibrium}}}{[\text{HNO}_2]_{\text{initial}}} \times 100\% = \frac{3.7 \times 10^{-3} \text{ M}}{0.035 \text{ M}} \times 100\% = 11\%$$

Weak Acids

Using K_a to Calculate pH

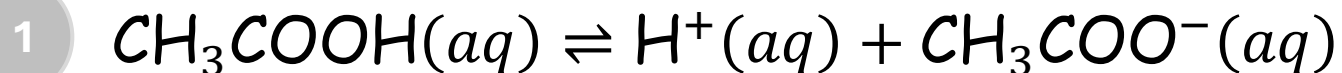
How to Calculate pH Using K_a ?

1. Write the ionization equilibrium.
2. Write the equilibrium expression and the value for the equilibrium constant.
3. Express the concentrations involved in the equilibrium reaction.
4. Substitute the equilibrium concentrations into the equilibrium-constant expression and solve for x.

Weak Acids

Using K_a to Calculate pH

Example: Calculate the pH at 25 °C of a 0.30 M solution of acetic acid.



2
$$K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = 1.8 \times 10^{-5}$$

(Table 16.2)

3

	$[\text{CH}_3\text{COOH}]$	$[\text{H}^+]$	$[\text{CH}_3\text{COO}^-]$
Initial (M)	0.03	0	0
Change (M)	-x	+x	+x
Equilibrium (M)	0.03-x	x	x

4
$$K_a = \frac{x^2}{0.30} = 1.8 \times 10^{-5}$$

$$x^2 = (0.30)(1.8 \times 10^{-5}) = 5.4 \times 10^{-6}$$

$$x = \sqrt{5.4 \times 10^{-6}} = 2.3 \times 10^{-3}$$

$$[\text{H}^+] = x = 2.3 \times 10^{-3} \text{ M}$$

$$\text{pH} = -\log(2.3 \times 10^{-3}) = 2.64$$

Q. From these data, calculate the percent ionization of the acid.

Weak Acids

- Q. Calculate the pH and percentage of HF molecules ionized in a 0.10 M HF solution ($K_a = 6.8 \times 10^{-4}$).
- Q. Calculate the pH of a 0.20 M solution of HCN. (Refer to Table 16.2 or Appendix D in the textbook for the value of K_a .)
- Q. What is the pH of a 0.010 M solution of HF?
- Q. What is the pH of a 0.40 M solution of benzoic acid, C_6H_5COOH ? (The K_a value for benzoic acid is given in Table 16.2.)
- Q. A solution of benzoic acid has a pH of 2.63 ($K_a = 6.3 \times 10^{-5}$). What is the concentration of the solution?

Weak Acids

Weak Acid Concentration Vs. Ionization

- Increasing the concentration of a **weak acid** increases the equilibrium concentration of H^+ .

Reaction proceeds more rapidly in strong acid, leading to formation of larger H_2 bubbles and rapid disappearance of metal.

Reaction is complete in strong acid.

H_2 bubbles show reaction still in progress in weak acid.

Reaction eventually goes to completion in both acids.

1 M $HCl(aq)$
 $[H^+] = 1 M$

1 M $CH_3COOH(aq)$
 $[H^+] = 0.004 M$

Strong acid:

$$\checkmark [H^+]_{eq} = [HA]_{init}$$

Weak acid:

$$\checkmark [H^+]_{eq} < [HA]_{init}$$

- This creates a difference in conductivity and in rates of chemical reactions.

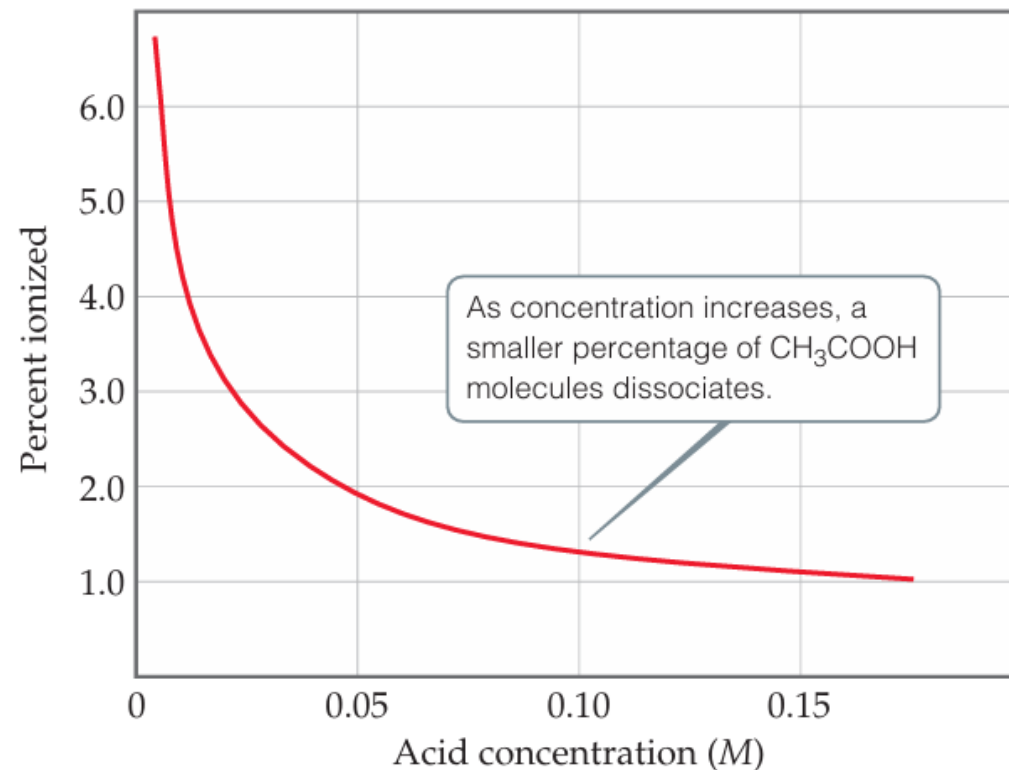
The same reaction has different rates in a weak acid solution compared to a strong acid solution.

Weak Acids

Weak Acid Concentration Vs. Ionization

- For a weak acid, the percent ionization decreases as acid concentration increases.
- Therefore, $[H^+]$ is not directly proportional to the weak acid concentration.

Example: Doubling the acid concentration does not double the H^+ concentration.



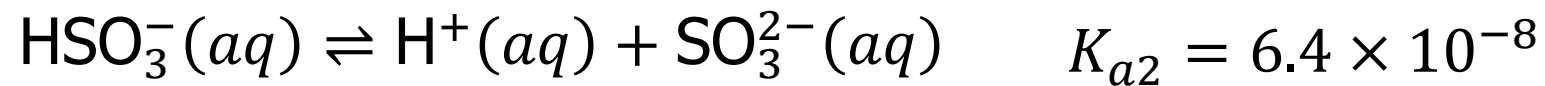
Effect of concentration on percent ionization in an acetic acid solution

Weak Acids

Polyprotic Acids

- Acids that have more than one ionizable H atom are known as polyprotic acids
- They undergo multi-step ionization.

Example : Sulfurous acid (H_2SO_3)



Q. Why is K_{a2} much lower than K_{a1} ?

- Each successive proton is harder to remove.
- Therefore, K_a decreases with each proton removed.
- $K_{a1} > K_{a2} > K_{a3}$

Weak Acids

Polyprotic Acids

TABLE 16.3 Acid-Dissociation Constants of Some Common Polyprotic Acids

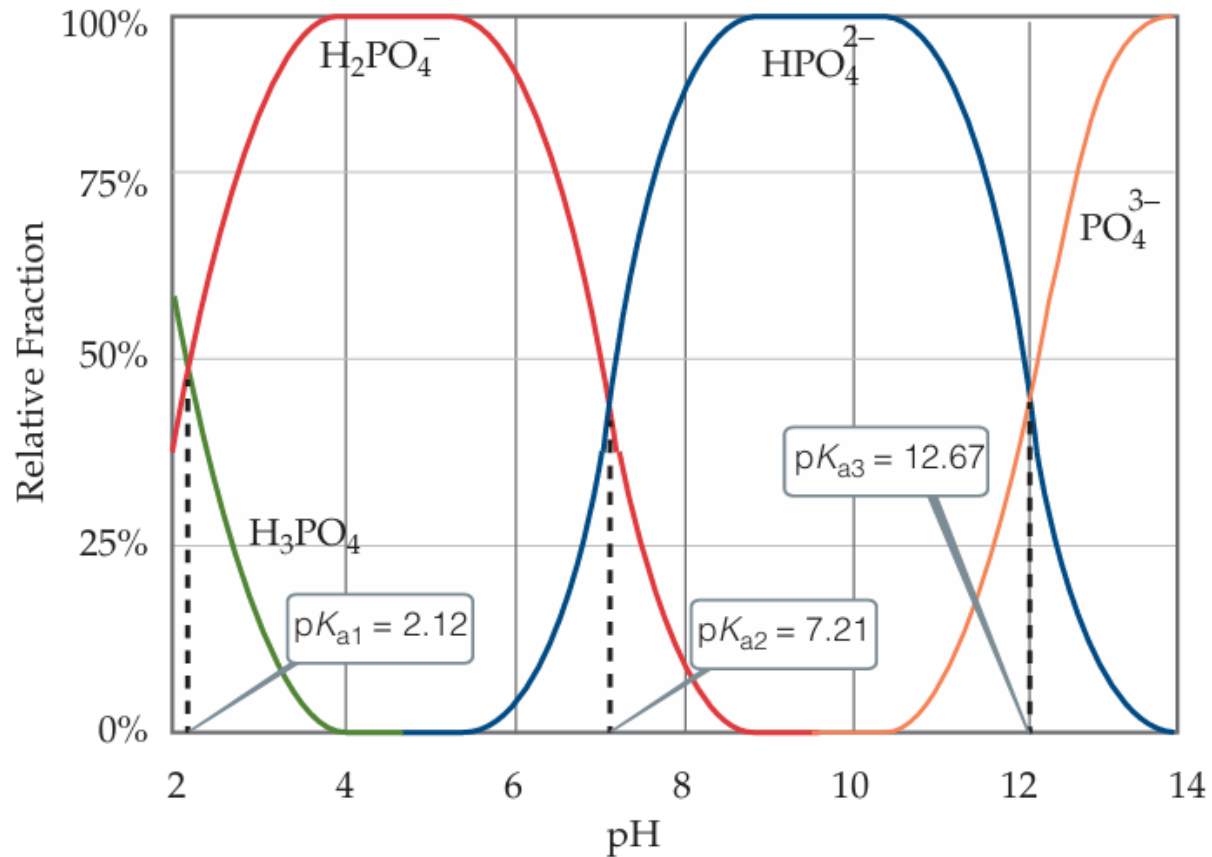
Name	Formula	K_{a1}	K_{a2}	K_{a3}
Ascorbic	$\text{H}_2\text{C}_6\text{H}_6\text{O}_6$	8.0×10^{-5}	1.6×10^{-12}	
Carbonic	H_2CO_3	4.3×10^{-7}	5.6×10^{-11}	
Citric	$\text{H}_3\text{C}_6\text{H}_5\text{O}_7$	7.4×10^{-4}	1.7×10^{-5}	4.0×10^{-7}
Oxalic	$\text{HOOC}-\text{COOH}$	5.9×10^{-2}	6.4×10^{-5}	
Phosphoric	H_3PO_4	7.5×10^{-3}	6.2×10^{-8}	4.2×10^{-13}
Sulfurous	H_2SO_3	1.7×10^{-2}	6.4×10^{-8}	
Sulfuric	H_2SO_4	Large	1.2×10^{-2}	
Tartaric	$\text{C}_2\text{H}_2\text{O}_2(\text{COOH})_2$	1.0×10^{-3}	4.6×10^{-5}	

Extended K_a value: Appendix D in the textbook

- ✓ If successive K_a values differ by $\geq 10^3$, pH can usually be estimated accurately.
- ✓ Treat the polyprotic acid as a monoprotic acid.
- ✓ Consider only the first dissociation (K_{a1}).

Weak Acids

Polyprotic Acids



- Q. Write the ionization process/steps of phosphoric acid.
- Q. Write the equilibrium constant expressions for K_{a2} and K_{a3} .

Relative fraction of phosphate species in water as a function of pH for phosphoric acid.

Weak Acids

Polyprotic Acids

- Q. The solubility of CO_2 in water at 25 °C and 10.13 kPa is 0.0037 M. The common practice is to assume that all the dissolved CO_2 is in the form of carbonic acid (H_2CO_3) which is produced in the reaction:



What is the pH of a 0.0037 M solution of H_2CO_3 ?

- Q. What is the pH of a 0.28 M solution of ascorbic acid (Vitamin C)? (See Table 16.3 for K_{a1} and K_{a2} .)

Weak Bases

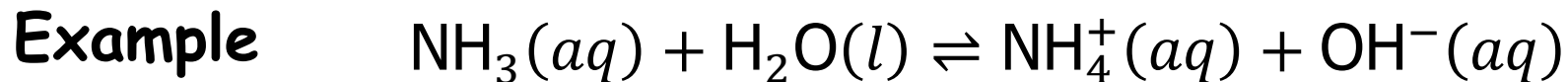
Q. How are weak bases different from strong bases?

- Weak bases react with water by accepting (abstracting) H^+ .
- This forms the conjugate acid of the base and OH^- ions.



$$K_b = \frac{[\mathbf{HB}^+][\mathbf{OH}^-]}{[\mathbf{B}]}$$

K_b : base-dissociation constant



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$$

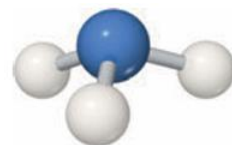
Weak Bases

TABLE 16.4 Some Weak Bases in Water at 25 °C

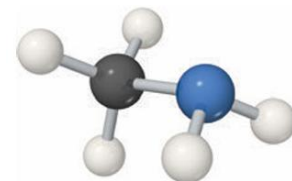
Base	Structural Formula*	Conjugate Acid	K_b
Ammonia (NH ₃)		NH ₄ ⁺	1.8×10^{-5}
Pyridine (C ₅ H ₅ N)		C ₅ H ₅ NH ⁺	1.7×10^{-9}
Hydroxylamine (HONH ₂)		HONH ₃ ⁺	1.1×10^{-8}
Methylamine (CH ₃ NH ₂)		CH ₃ NH ₃ ⁺	4.4×10^{-4}
Hydrosulfide ion (HS ⁻)		H ₂ S	1.8×10^{-7}
Carbonate ion (CO ₃ ²⁻)		HCO ₃ ⁻	1.8×10^{-4}
Hypochlorite ion (ClO ⁻)		HClO	3.3×10^{-7}

*The atom that accepts the proton is shown in blue.

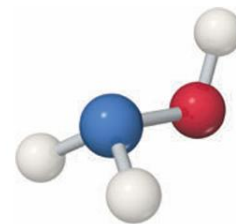
- Q. Identify the strongest base from this table.
- Q. Identify the weakest base from this table.
- Q. When hydroxylamine acts as a base, which atom accepts the proton?
- Q. Identify the strongest conjugate acid from this table.



Ammonia
NH₃



Methylamine
CH₃NH₂

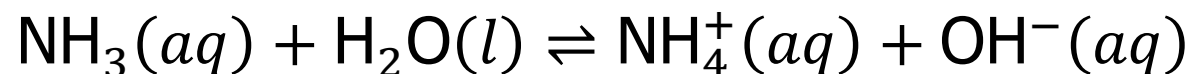


Hydroxylamine
NH₂OH

Weak Bases

Q. How to calculate concentration Using K_b value?

Q. Calculate the concentration of OH^- in a 0.15 M solution of NH_3



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]} = 1.8 \times 10^{-5}$$

$$K_b = \frac{(x)(x)}{0.15 - x} = 1.8 \times 10^{-5}$$

$$\frac{x^2}{0.15} = 1.8 \times 10^{-5}$$

$$x^2 = (0.15)(1.8 \times 10^{-5}) = 2.7 \times 10^{-6}$$

$$x = [\text{NH}_4^+] = [\text{OH}^-]$$

$$x = \sqrt{2.7 \times 10^{-6}} = 1.6 \times 10^{-3} \text{ M}$$

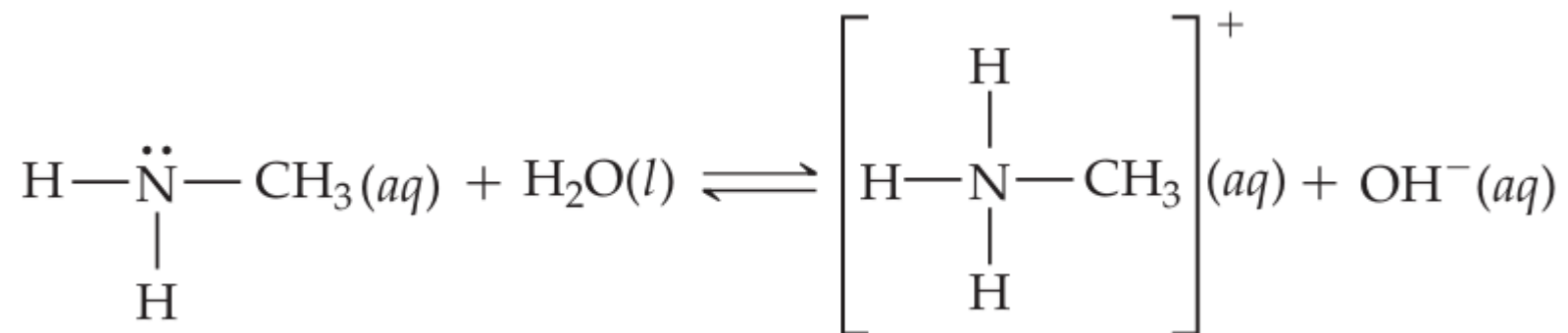
	NH_3	H_2O	NH_4^+	OH^-
Initial (M)	0.15	-	0	0
Change (M)	-x	-	+x	+x
Equilibrium (M)	0.15-x	-	x	x

Weak Bases

Types of weak bases

Neutral weak bases (See table 16.4)

- Contain an atom with a lone pair of electrons that can accept H^+ .
- Most are nitrogen-containing compounds, such as ammonia (NH_3) and **amines**.
 - ✓ In amines, one or more N–H bonds in NH_3 are replaced by N–C bonds.
 - ✓ Amines accept H^+ from water by forming a new N–H bond.
- Example: Methylamine (CH_3NH_2) acts as a weak base.

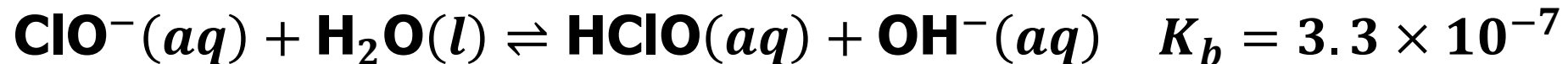


Weak Bases

Types of weak bases

Anions of weak acids

- Example: $\text{NaClO} \rightarrow \text{Na}^+ + \text{ClO}^-$
- Na^+ is a spectator ion and does not participate in the acid–base reaction.
- ClO^- is the conjugate base of the weak acid HClO .
- Therefore, ClO^- acts as a **weak base** in water by accepting H^+ .

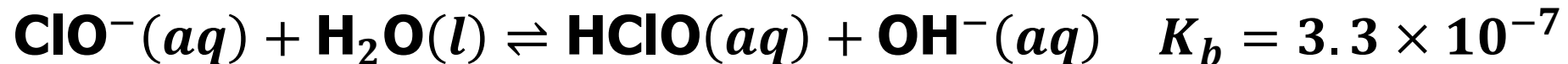


Weak Bases

Types of weak bases

Anions of weak acids

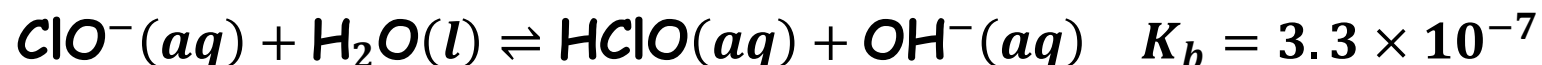
- Example: $\text{NaClO} \rightarrow \text{Na}^+ + \text{ClO}^-$
- Na^+ is a spectator ion and does not participate in the acid–base reaction.
- ClO^- is the conjugate base of the weak acid HClO .
- Therefore, ClO^- acts as a **weak base** in water by accepting H^+ .



Weak Bases

Using pH to Determine the Concentration of a Salt

- Q. A solution made by adding solid sodium hypochlorite (NaClO) to enough water to make 2.00 L of solution has a pH of 10.50.



Calculate the number of moles of NaClO added to the water.

$$\checkmark \quad pOH = 14.00 - pH = 14.00 - 10.50 = 3.50$$

$$\checkmark \quad [\text{OH}^{-}] = 10^{-3.50} = 3.2 \times 10^{-4} \text{ M}$$

	ClO^{-}	H_2O	HClO	OH^{-}
Initial (M)	$\times$	-	0	0
Change (M)	-3.2×10^{-4}	-	$+3.2 \times 10^{-4}$	$+3.2 \times 10^{-4}$
Equilibrium (M)	$x - 3.2 \times 10^{-4}$	-	3.2×10^{-4}	3.2×10^{-4}

$$K_b = \frac{[\text{HClO}][\text{OH}^{-}]}{[\text{ClO}^{-}]}$$

$$= \frac{(3.2 \times 10^{-4})^2}{x - 3.2 \times 10^{-4}} = 3.3 \times 10^{-7}$$

$$x = \frac{(3.2 \times 10^{-4})^2}{3.3 \times 10^{-7}} + (3.2 \times 10^{-4})$$

$$\mathbf{x = 0.31 \text{ M}}$$

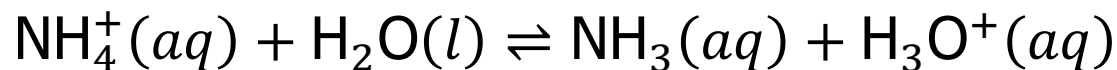
Weak Bases

- Q. The benzoate ion, $\text{C}_6\text{H}_5\text{COO}^-$, is a weak base with $K_b = 1.6 \times 10^{-10}$. How many moles of sodium benzoate are present in 0.50 L of a solution of $\text{NaC}_6\text{H}_5\text{COO}$ if the pH is 9.04?
- Q. Calculate the molar concentration of OH^- in a 0.724 M solution of hypobromite ion (BrO^- ; $K_b = 4.0 \times 10^{-6}$). What is the pH of this solution?

Weak Bases

Relationship Between K_a and K_b

- The stronger an acid, the weaker its conjugate base



1



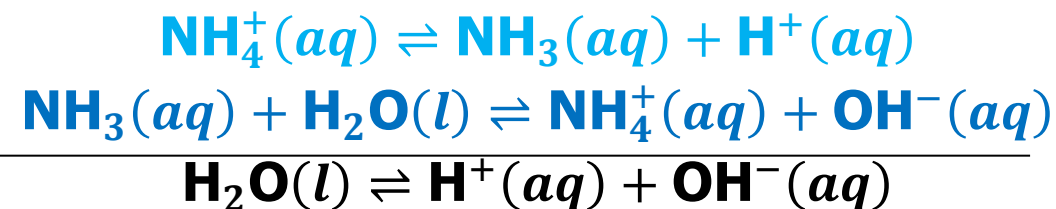
$$K_a = \frac{[\text{NH}_3][\text{H}^+]}{[\text{NH}_4^+]}$$

2



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$$

3



$$[\text{H}^+][\text{OH}^-] = K_w$$

$$K_a \times K_b = \left(\frac{[\text{NH}_3][\text{H}^+]}{[\text{NH}_4^+]} \right) \left(\frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]} \right)$$

4

$$K_a \times K_b = K_w$$

(For a conjugate acid-base pair)

Weak Bases

Relationship Between K_a and K_b

TABLE 16.5 Some Conjugate Acid–Base Pairs

Acid	K_a	Base	K_b
HNO ₃	(Strong acid)	NO ₃ [−]	(Negligible basicity)
HF	6.8×10^{-4}	F [−]	1.5×10^{-11}
CH ₃ COOH	1.8×10^{-5}	CH ₃ COO [−]	5.6×10^{-10}
H ₂ CO ₃	4.3×10^{-7}	HCO ₃ [−]	2.3×10^{-8}
NH ₄ ⁺	5.6×10^{-10}	NH ₃	1.8×10^{-5}
HCO ₃ [−]	5.6×10^{-11}	CO ₃ ^{2−}	1.8×10^{-4}
OH [−]	(Negligible acidity)	O ^{2−}	(Strong base)

Extended K_a value: Appendix D in the textbook

Q. For each of **conjugate-acid base** pairs in the above table, confirm that $K_a \times K_b = 1.0 \times 10^{-14}$.

Weak Bases

Relationship Between K_a and K_b

$$\text{pH} = -\log[\text{H}^+]$$

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{p}K_w = -\log K_w$$

$$\text{p}K_a = -\log K_a$$

$$K_a \times K_b = K_w$$

$$\text{p}K_a + \text{p}K_b = \text{p}K_w$$

$$\text{p}K_a + \text{p}K_b = 14$$

At 25 °C for any conjugate acid-base pair.

Q. Consider the following table:

Species	CH_3COOH	HCN	HNO_2	NH_4^+
$\text{p}K_a$	4.74	9.31	3.35	9.26

Write the conjugate bases of these four weak acids and rank them in order of increasing base strength



Weak Bases

- Q. Calculate (see table 16.5):
- K_b for the fluoride ion
 - K_a for the ammonium ion
- Q. Codeine ($C_{18}H_{21}NO_3$) is a weak organic base. A 5.0×10^{-3} M solution of codeine has a pH of 9.95. Calculate the value of K_b for this substance. What is the pK_b for this base?
- Q. Phenol, C_6H_5OH , has a K_a of 1.3×10^{-10} .
- Write out the K_a reaction for phenol.
 - Calculate K_b for phenol's conjugate base.
 - Is phenol a stronger or weaker acid than water?

Acid–Base Properties of Salt Solutions

Acid–Base Properties of Salt Solutions

- Salts are strong electrolytes and completely dissociate in water.
- Acid–base properties of solutions depend on the behavior of individual cations and anions.
- Many ions react with water to generate H^+ (*aq*) or $[\text{OH}^-]$ (*aq*) ions in a process called **hydrolysis**.
- The pH of an aqueous salt solution is predicted qualitatively by analyzing its constituent cations and anions.

Acid–Base Properties of Salt Solutions

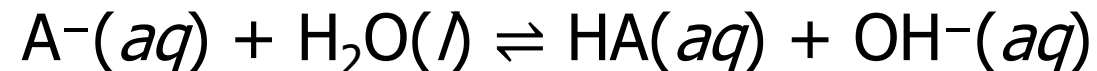
An Anion's Ability to React with Water

- An anion A^- is generally the conjugate base of an acid HA . To identify HA , add H^+ to A^- .

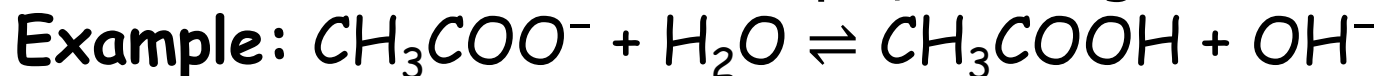
1. If HA is a strong acid (is among the seven strong acids), A^- has negligible basicity and does not react significantly with water.

Example: Cl^- do not produce OH^- from water or affect pH and is considered a spectator ion.

2. If HA is a weak acid (not one of the seven common strong acids), A^- acts as a weak base and reacts slightly with water:



- Formation of OH^- increases pH, making the solution basic.



Acid–Base Properties of Salt Solutions

An Anion's Ability to React with Water

3. Some anions (obtained from salts) contain ionizable protons and are **amphiprotic** (they can act as either an acid or a base).

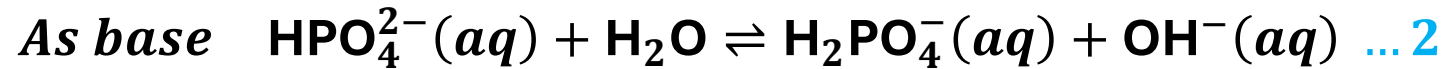
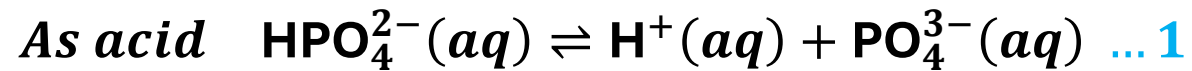
✓ Example: HSO_3^-

- The effect of an amphiprotic ion on solution pH depends on the relative values of K_a and K_b :
 - ✓ $K_a > K_b \rightarrow$ acidic solution
 - ✓ $K_b > K_a \rightarrow$ basic solution

Acid–Base Properties of Salt Solutions

An Anion's Ability to React with Water

Q. Predict whether the salt Na_2HPO_4 forms an acidic solution or a basic solution when dissolved in water.



Of these two reactions, the one with the larger equilibrium constant determines whether the solution is acidic or basic.

- ✓ The value of K_a (Equation 1) is K_{a3} for H_3PO_4 : 4.2×10^{-13} (Table 16.3)
- ✓ Calculate K_b (Equation 2) from K_a of its conjugate acid (H_2PO_4^-) and the relationship $K_a \times K_b = K_w$

K_a for H_2PO_4^- is K_{a2} for H_3PO_4 : 6.2×10^{-8} (from Table 16.3)

$$K_b(\text{HPO}_4^{2-}) \times K_a(\text{H}_2\text{PO}_4^-) = K_w = 1.0 \times 10^{-14}$$

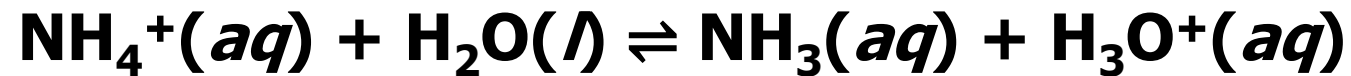
$$K_b(\text{HPO}_4^{2-}) = \frac{1.0 \times 10^{-14}}{6.2 \times 10^{-8}} = 1.6 \times 10^{-7}$$

$K_b > K_a$ (the solution is basic)

Acid–Base Properties of Salt Solutions

A Cation's Ability to React with Water

1. Polyatomic cations containing H^+ can act as the conjugate acids of weak bases.
 - Example: NH_4^+ is the conjugate acid of the weak base NH_3 .



- NH_4^+ donates a proton to water
- Formation of H_3O^+ lowers the pH, making the solution acidic.

Acid–Base Properties of Salt Solutions

A Cation's Ability to React with Water

2. Some small, highly charged metal cations, such as Fe^{3+} , also produce acidic solutions.
 - In $\text{Fe}(\text{NO}_3)_3$, the acidity is not caused by NO_3^- because nitrate is the conjugate base of the strong acid HNO_3 and has negligible reaction with water.
 - The Fe^{3+} ion is responsible for the acidic behavior.
 - The ions of alkali and alkaline earth metals do not react with water and therefore do not affect pH.

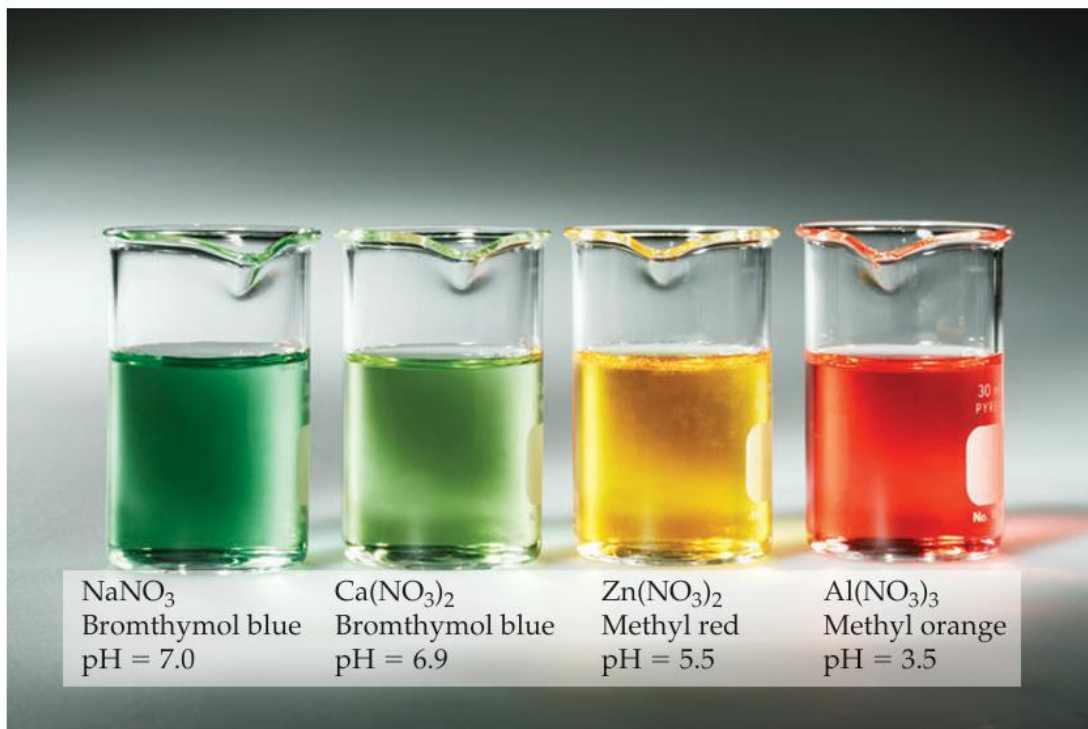
TABLE 16.6 Acid-Dissociation Constants for Metal Cations in Aqueous Solution at 25 °C

Cation	K_a
Fe^{3+}	6.3×10^{-3}
Cr^{3+}	1.6×10^{-4}
Al^{3+}	1.4×10^{-5}
Fe^{2+}	3.2×10^{-10}
Zn^{2+}	2.5×10^{-10}
Ni^{2+}	2.5×10^{-11}

Acid–Base Properties of Salt Solutions

A Cation's Ability to React with Water

2. Some small, highly charged metal cations, such as Fe^{3+} , also produce acidic solutions.



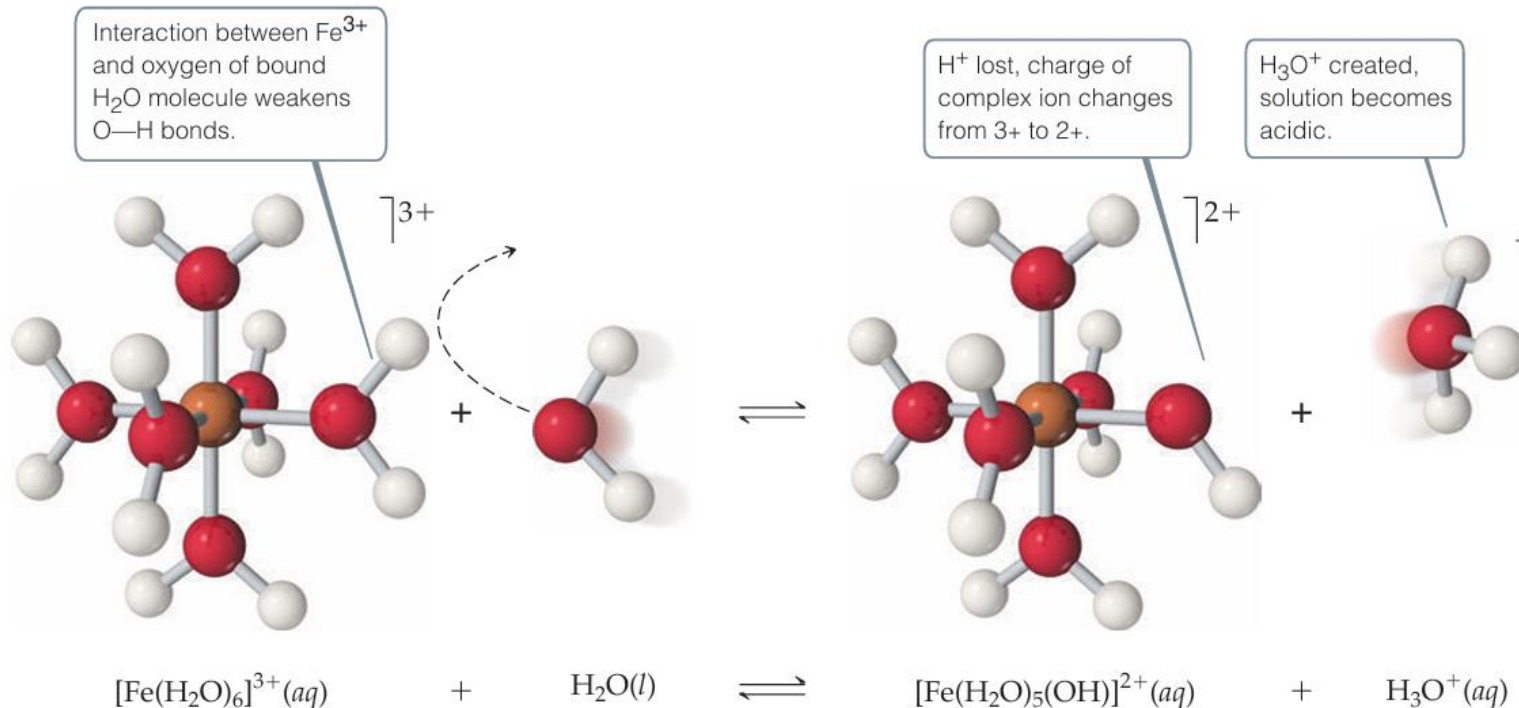
▲ **Figure 16.17** Effect of cations on solution pH. The pH values of 1.0 M solutions of four nitrate salts are estimated using acid–base indicators.

- ✓ Group I or group II metal cations are neutral.
- ✓ Polyatomic cations are typically the conjugate acids of a weak base; for example, NH_4^+ .
- ✓ Transition and post-transition metal cations are acidic. Why?

Acid–Base Properties of Salt Solutions

A Cation's Ability to React with Water

2. Some small, highly charged metal cations, such as Fe^{3+} , also produce acidic solutions.



▲ **Figure 16.18** A hydrated Fe^{3+} ion acts as an acid by donating an H^+ from a bound water molecule to a free H_2O molecule, forming H_3O^+ .

- ✓ Metal cations become surrounded by water molecules through **hydration**.
- ✓ Because metal ions are positively charged, they attract the unshared electron pairs on oxygen in water molecules.
- ✓ Higher charge and smaller size
→ stronger attraction to water
→ greater acidity.

Acid–Base Properties of Salt Solutions

Combined Effect of Cation and Anion in Solution

Case	Cation behavior	Anion behavior	Expected pH	Mechanism	Examples
1	Does not react with water	Does not react with water	Neutral	Neither H_3O^+ nor OH^- is produced significantly	NaCl , $\text{Ba}(\text{NO}_3)_2$, RbClO_4
2	Does not react with water	Produces OH^-	Basic	OH^- increases the pH	NaClO , RbF , BaSO_3
3	Produces H_3O^+	Does not react with water	Acidic	H_3O^+ decreases the pH	NH_4NO_3 , AlCl_3 , $\text{Fe}(\text{NO}_3)_3$
4	Produces H_3O^+	Produces OH^-	Acidic, Basic, or Neutral	Depends on the relative strengths of the cation and anion	NH_4ClO , $\text{Al}(\text{CH}_3\text{COO})_3$, CrF_3

NaCl : Sodium chloride; $\text{Ba}(\text{NO}_3)_2$: Barium nitrate; RbClO_4 : Rubidium perchlorate; NaClO : Sodium hypochlorite; RbF : Rubidium fluoride; BaSO_3 : Barium sulfite; NH_4NO_3 : Ammonium nitrate; AlCl_3 : Aluminum chloride; $\text{Fe}(\text{NO}_3)_3$: Iron(III) nitrate; NH_4ClO : Ammonium hypochlorite; $\text{Al}(\text{CH}_3\text{COO})_3$: Aluminum acetate; CrF_3 : Chromium(III) fluoride.

Acid–Base Properties of Salt Solutions

- Q.** How many of the following salts are expected to produce acidic solutions: NaHSO_4 , NaHC_2O_4 , NaH_2PO_4 , and NaHCO_3 ? (hint: see Table 16.3 for data)
- Q.** Determine whether aqueous solutions of each of these salts are acidic, basic, or neutral: (a) $\text{Ba}(\text{CH}_3\text{COO})_2$, (b) NH_4Cl , (c) $\text{CH}_3\text{NH}_3\text{Br}$, (d) KNO_3 , (e) $\text{Al}(\text{ClO}_4)_3$, (f) AlCl_3 , (g) NaBr , (h) NaClO , (i) $[\text{CH}_3\text{NH}_3]\text{NO}_3$, (j) Na_2SO_3
- Q.** Order the following solutions from lowest to highest pH:
(i) 0.10 M NaClO , (ii) 0.10 M KBr , (iii) 0.10 M NH_4ClO_4
- Q.** A dilute solution of a salt had the measured pH of 8.3. Which one of these salts is likely to be in the solution? (a) AlCl_3 (b) NaBr (c) NH_4NO_3 (d) K_2CO_3 (e) $\text{CH}_3\text{NH}_3\text{Cl}$

NaHSO_4 : Sodium hydrogen sulfate (Sodium bisulfate); NaHC_2O_4 : Sodium hydrogen oxalate (Sodium bioxalate); NaH_2PO_4 : Sodium dihydrogen phosphate; NaHCO_3 : Sodium hydrogen carbonate (Sodium bicarbonate); $\text{Ba}(\text{CH}_3\text{COO})_2$: Barium acetate; NH_4Cl : Ammonium chloride; $\text{CH}_3\text{NH}_3\text{Br}$: Methylammonium bromide; KNO_3 : Potassium nitrate; $\text{Al}(\text{ClO}_4)_3$: Aluminum perchlorate; AlCl_3 : Aluminum chloride; NaBr : Sodium bromide; NaClO : Sodium hypochlorite; $[\text{CH}_3\text{NH}_3]\text{NO}_3$: Methylammonium nitrate; Na_2SO_3 : Sodium sulfite; KBr : Potassium bromide; NH_4ClO_4 : Ammonium perchlorate; NH_4NO_3 : Ammonium nitrate; K_2CO_3 : Potassium carbonate; $\text{CH}_3\text{NH}_3\text{Cl}$: Methylammonium chloride

Acid–Base Behavior and Chemical Structure

Factors That Affect Acid Strength

- The strength of an acid is often affected by three separate factors.

Factor	What Determines It?	Effect on Acid Strength	Key Idea (Example)
1. H–A Bond Polarity 	How strongly the H–A bond pulls electron density away from H	Greater polarity → stronger acid	H becomes more δ^+ and is easier to donate as H^+
2. H–A Bond Strength	How strongly H is bonded to A	Weaker bond → stronger acid	HF has a very strong H–F bond → weak acid; HCl, HBr, and HI have weaker bonds → stronger acids
3. Conjugate Base Stability	How stable the species A^- is after H^+ is removed	Greater conjugate-base stability → stronger acid	Stable A^- makes proton loss more favorable

Q. Determine the acidity/basicity nature of NaH and CH_4 in water.

Acid–Base Behavior and Chemical Structure

Binary Acids

14	15	16	17
CH_4 Neither acid nor base	NH_3 Weak base $K_b = 1.8 \times 10^{-5}$	H_2O	HF Weak acid $K_a = 6.8 \times 10^{-4}$
SiH_4 Neither acid nor base	PH_3 Very weak base $K_b = 4 \times 10^{-28}$	H_2S Weak acid $K_a = 9.5 \times 10^{-8}$	HCl Strong acid
		H_2Se Weak acid $K_a = 1.3 \times 10^{-4}$	HBr Strong acid

Increasing acid strength

Increasing acid strength

Down a Group → Bond Strength Controls

- H–A bond strength decreases as atom A becomes larger.
- Weaker H–A bond → easier H^+ release → stronger acid

Examples: **HCl** > **HF**; H_2S > H_2O

Across a Period → Bond Polarity Controls

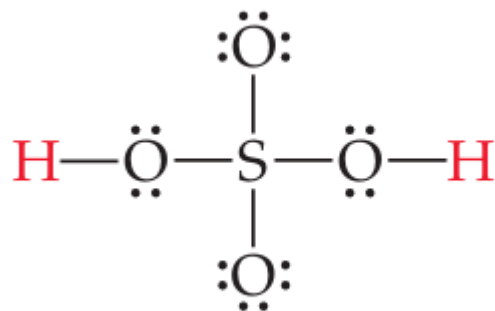
- Electronegativity of A generally increases from left to right.
- Greater H–A bond polarity → greater tendency to release H^+ → stronger acid.
- Period 2 trend: **$\text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O} < \text{HF}$**

Trends in acid strength for the binary hydrides of Periods 2–4

Acid–Base Behavior and Chemical Structure

Oxyacids

- Acids in which OH groups and possibly additional oxygen atoms are bound to a central atom are called **oxyacids**.

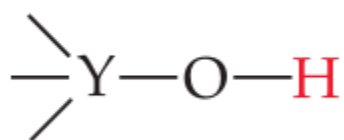


Q. What factors determine whether a given OH group behaves as a base or as an acid?

Acid–Base Behavior and Chemical Structure

Oxyacids

Q. What factors determine whether a given OH group behaves as a base or as an acid?

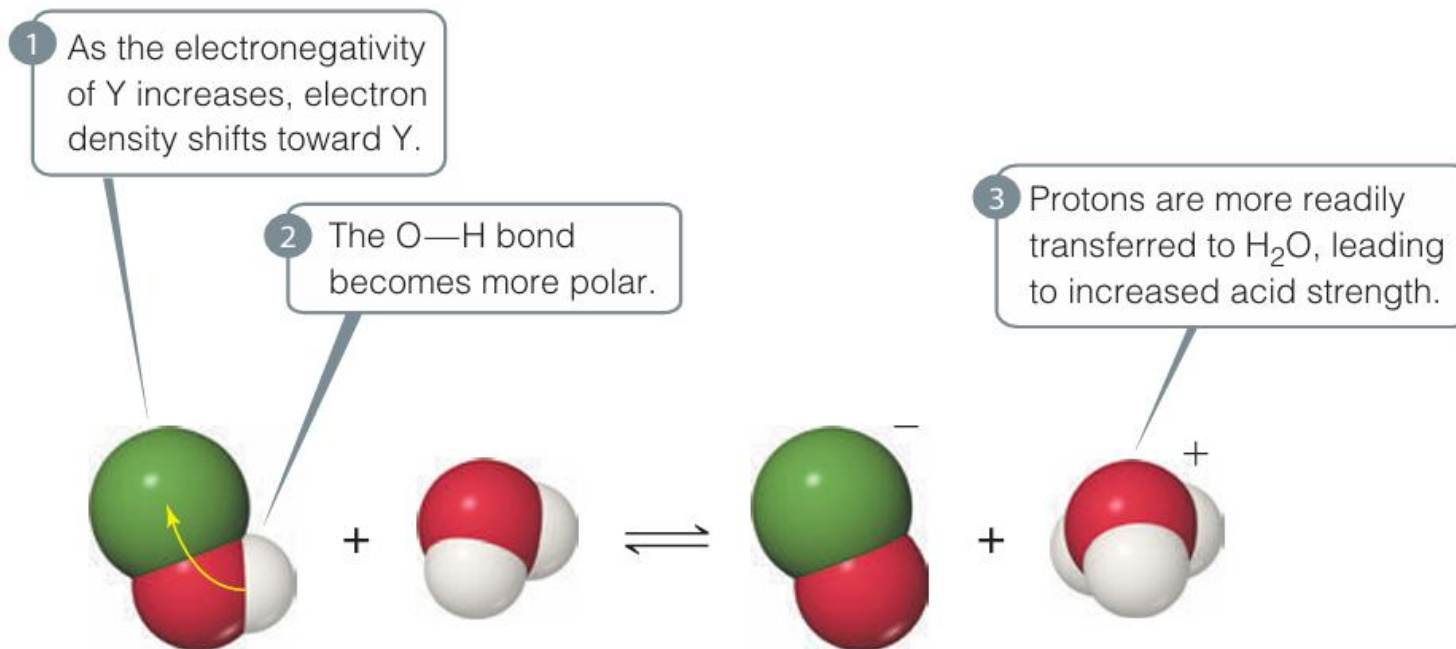


- Consider an OH group bound to some atom Y, which might in turn have other groups attached to it.

Y	Type	Bond with O	Main behavior	Effect of increasing electronegativity of Y
If metal (Na, Mg)	Ionic	Electrons transferred to O	Forms OH ⁻ → Basic	Not the main trend
If nonmetal	Covalent	Electrons shared	Usually acidic or neutral	Acidity increases

Acid–Base Behavior and Chemical Structure

Oxyacids



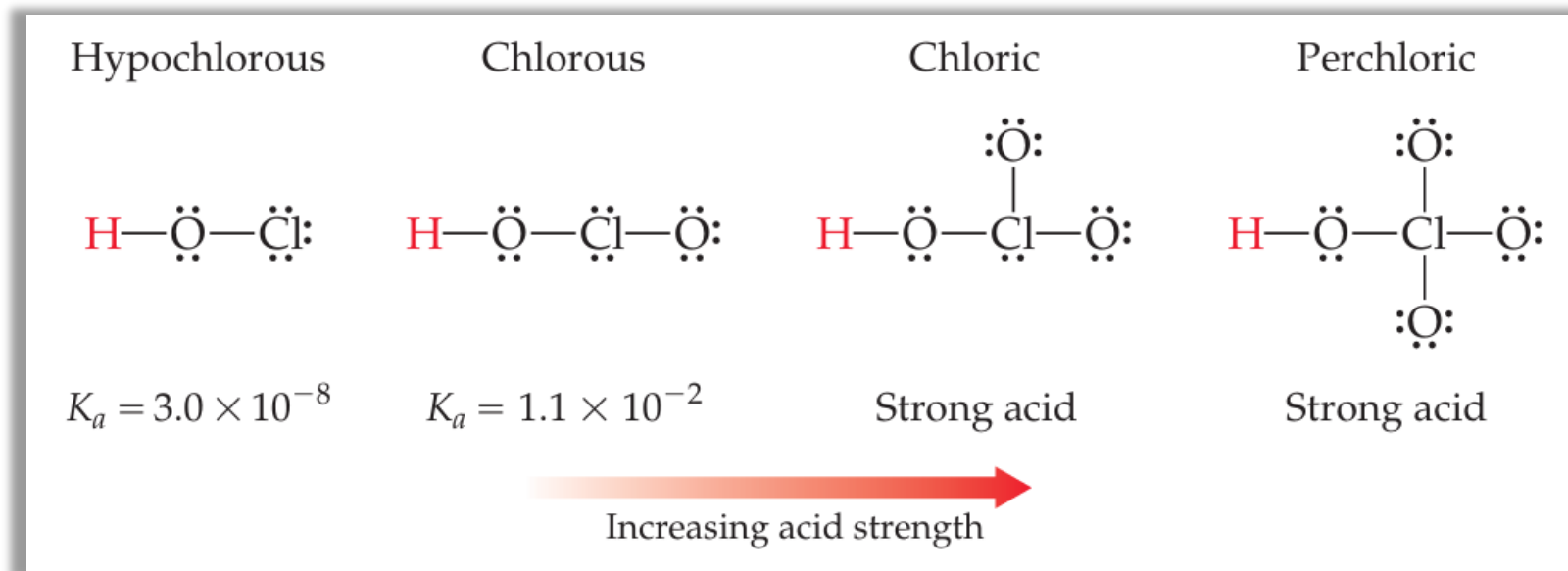
Substance	Y—OH	Electronegativity of Y	Dissociation constant
Hypochlorous acid	Cl—OH	3.0	$K_a = 3.0 \times 10^{-8}$
Hypobromous acid	Br—OH	2.8	$K_a = 2.5 \times 10^{-9}$
Hypoiodous acid	I—OH	2.5	$K_a = 2.3 \times 10^{-11}$
Water	H—OH	2.1	$K_w = 1.0 \times 10^{-14}$

Acidity of the hypohalous oxyacids (YOH) and water as a function of the electronegativity of Y.

Acid–Base Behavior and Chemical Structure

Oxyacids

- Many oxyacids contain additional oxygen atoms bonded to the central atom Y.
- The strength of an acid increases as additional electronegative atoms bond to the central atom Y
- In a series of oxyacids, the acidity increases as the oxidation number of the central atom increases.



Strength of the chlorine oxyacids.

Acid–Base Behavior and Chemical Structure

Q. Arrange the compounds in each series in order of increasing acid strength.

a. AsH_3 , HBr , KH , H_2Se

b. H_2SO_4 , H_2SeO_3 , H_2SeO_4

Q. Which of the following shows the correct order of acid strength?

a. $\text{HIO}_2 < \text{HOI} < \text{HClO}_3 < \text{HBrO}_2 < \text{HClO}_2$

b. $\text{HOI} < \text{HIO}_2 < \text{HBrO}_2 < \text{HClO}_2 < \text{HClO}_3$

c. $\text{HBrO}_2 < \text{HIO}_2 < \text{HClO}_2 < \text{HOI} < \text{HClO}_3$

d. $\text{HClO}_3 < \text{HClO}_2 < \text{HBrO}_2 < \text{HIO}_2 < \text{HOI}$

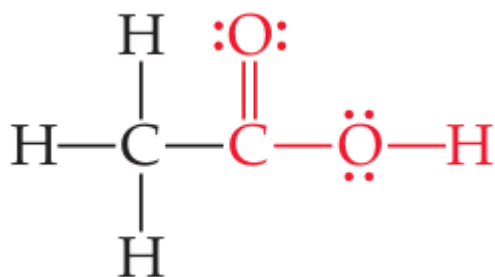
e. $\text{HOI} < \text{HClO}_2 < \text{HBrO}_2 < \text{HIO}_2 < \text{HClO}_3$

AsH_3 : Arsine; HBr : Hydrobromic acid; KH : Potassium hydride; H_2Se : Hydroselenic acid; H_2SO_4 : Sulfuric acid; H_2SeO_3 : Selenous acid; H_2SeO_4 : Selenic acid; HIO_2 : Iodous acid; HOI : Hypoiodous acid; HClO_3 : Chloric acid; HBrO_2 : Bromous acid; HClO_2 : Chlorous acid.

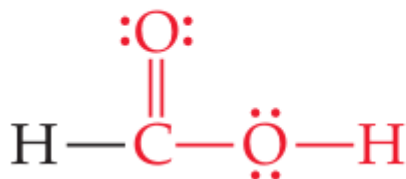
Acid–Base Behavior and Chemical Structure

Carboxylic Acids (A Major Class of Organic Acids)

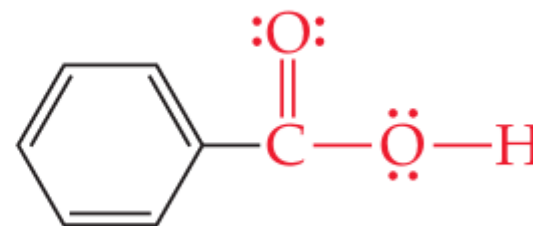
- Carboxylic acids contain the carboxyl group ($-\text{COOH}$).
- Only the H in $-\text{COOH}$ is readily ionized as H^+ .
- General form: $\text{R}-\text{COOH}$



Acetic acid



Formic acid



Benzoic acid

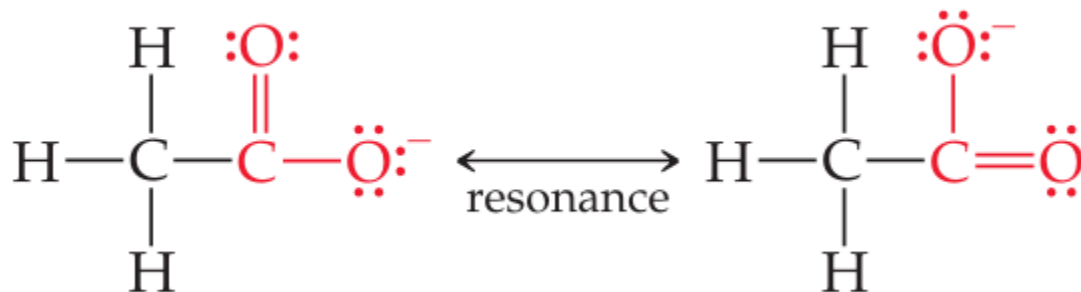
Q. What property makes them acidic?

Acid–Base Behavior and Chemical Structure

Carboxylic Acids (A Major Class of Organic Acids)

Q. What property makes them acidic?

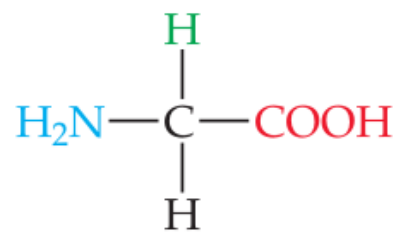
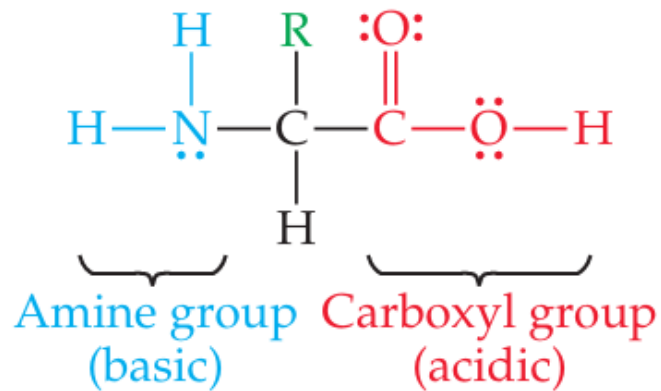
- 1. Polarity:** The second O atom pulls electron density away from the O–H bond → more polar O–H bond → easier H⁺ loss.
- 2. Resonance stabilization:** The carboxylate conjugate base (R–COO[–]) spreads its negative charge over two O atoms, increasing its stability.



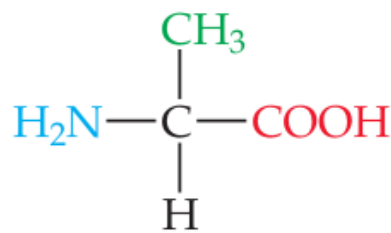
Acid–Base Behavior and Chemical Structure

The Amphiprotic Behavior of Amino Acids

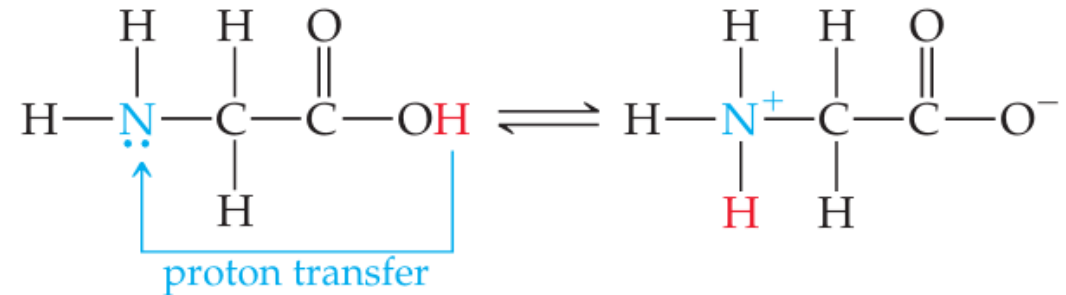
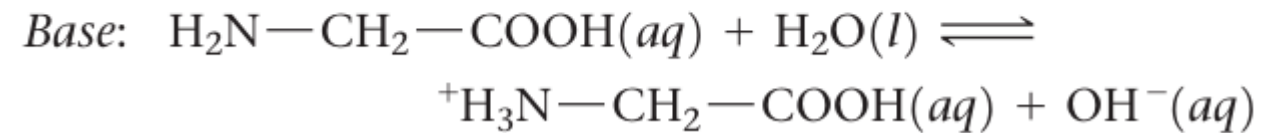
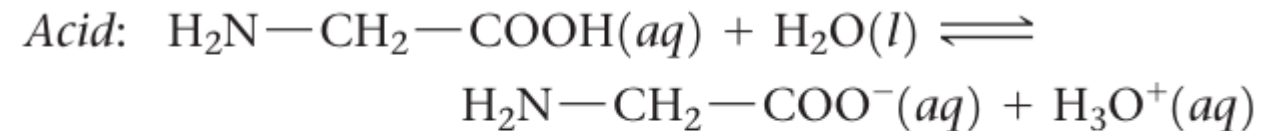
- Amino acids are the building blocks of proteins



Glycine



Alanine



Neutral molecule

Zwitterion

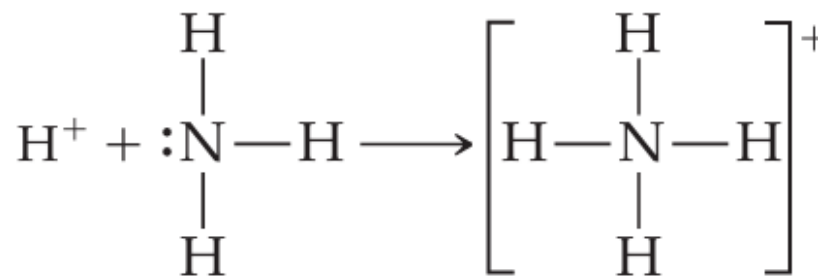
Acid–Base Behavior and Chemical Structure

Lewis Acids and Bases

Q. What property makes ammonia accept a proton (i.e act as a base)?

Brønsted–Lowry concept

- Base = proton (H^+) acceptor
- Acid = proton (H^+) donor



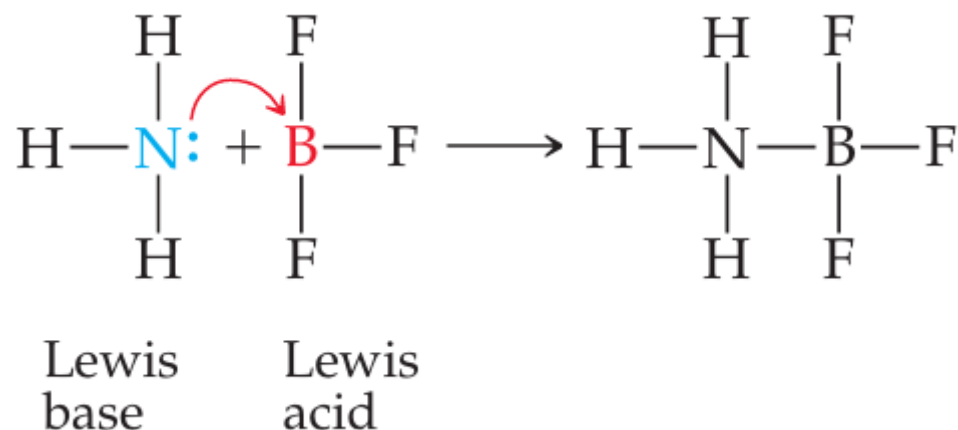
Lewis concept

- A Lewis acid is an electron-pair acceptor.
- A Lewis base is an electron-pair donor.
- Every Brønsted–Lowry base is also a **Lewis base** because accepting H^+ requires donating an electron pair.

Acid–Base Behavior and Chemical Structure

Lewis Acids and Bases

- Extends acid–base chemistry to reactions that do not involve proton transfer.
- Provides a broader description of chemical reactions.



NH_3 : Lewis Base

- Nitrogen has a lone pair.
- Donates the electron pair.

BF_3 : Lewis Acid

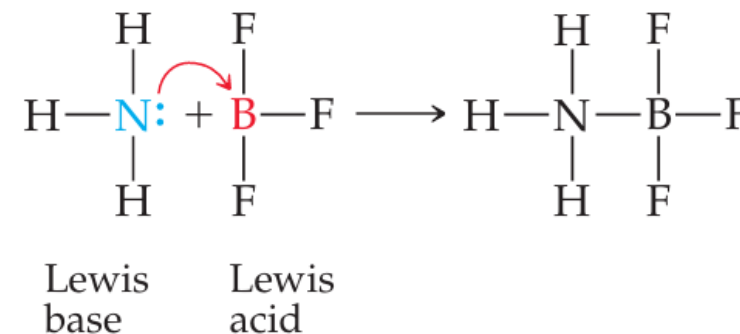
- Boron has an incomplete octet.
- Has a vacant orbital that can accept an electron pair.

Acid–Base Behavior and Chemical Structure

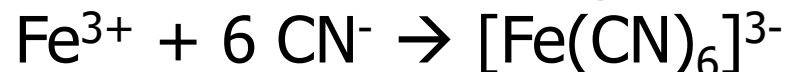
Lewis Acids and Bases: Common Lewis Acids

1. Molecules with incomplete octets

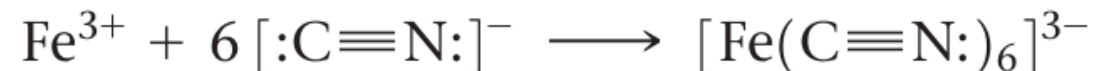
- ✓ In BF_3 , electron-deficient B accepts an electron pair.



2. Metal cations (having vacant orbitals)



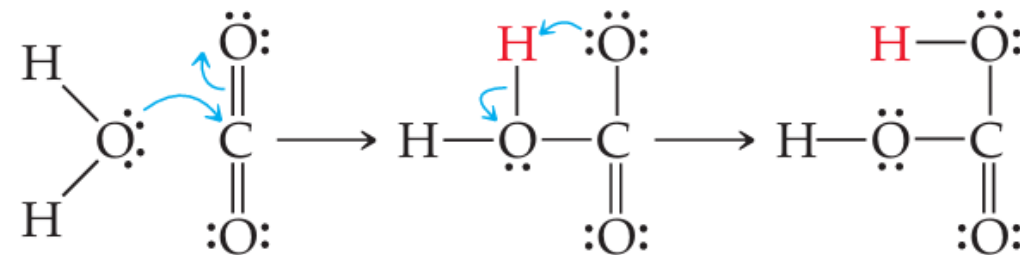
- ✓ Fe^{3+} = Lewis acid
- ✓ CN^- = Lewis base



3. Molecules with multiple bonds



- ✓ H_2O donates an electron pair.
- ✓ CO_2 accepts the electron pair.



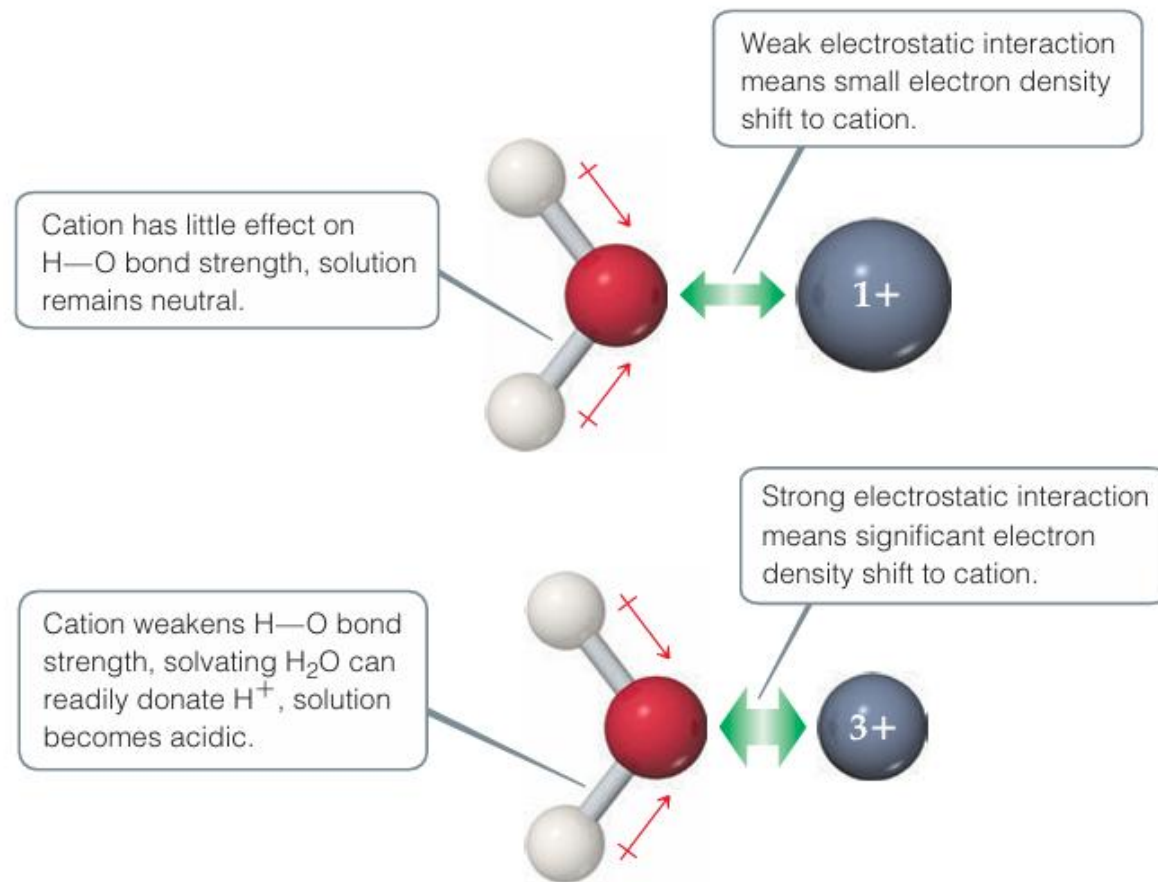
Acid–Base Behavior and Chemical Structure

Lewis Acids and Bases: Acidity of hydrated cation

- A metal cation acts as a Lewis acid by accepting electron density from **H₂O**



- ✓ The metal ion attracts electron density from O
- ✓ O–H bond becomes more polarized
- ✓ H⁺ is released more easily
- ✓ Acidity of coordinated H₂O increases
- This effect becomes more pronounced as the charge of the cation increases



The acidity of a hydrated cation depends on cation charge.

Summary Q

- Q. Consider Phosphorous acid (H_3PO_3).
- Draw the Lewis structure and identify the acidic hydrogens (diprotic or triprotic).
 - A 25.0 mL sample of an H_3PO_3 solution titrated with 0.102 M NaOH requires 23.3 mL of NaOH to neutralize both acidic protons. What is the molarity of the H_3PO_3 solution?
 - The original solution from part (b) has a pH of 1.59. Calculate the percent ionization and K_{a1} for H_3PO_3 , assuming that $K_{a1} \gg K_{a2}$.
 - How does the osmotic pressure of a 0.050 M solution of HCl compare qualitatively with that of a 0.050 M solution of H_3PO_3 ? Explain.
- Q. Identify the Lewis acid and Lewis base in each of the following reactions:
- $\text{HNO}_2(\text{aq}) + \text{OH}^-(\text{aq}) \rightleftharpoons \text{NO}_2^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 - $\text{FeBr}_3(\text{s}) + \text{Br}^-(\text{aq}) \rightleftharpoons \text{FeBr}_4^-(\text{aq})$
 - $\text{Zn}^{2+}(\text{aq}) + 4\text{NH}_3(\text{aq}) \rightleftharpoons \text{Zn}(\text{NH}_3)_4^{2+}(\text{aq})$
 - $\text{SO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{SO}_3(\text{aq})$

By now, you should be able to:

- ✓ Define and identify Arrhenius acids and bases.
- ✓ Define and identify Brønsted-Lowry acids and bases and identify conjugate acid-base pairs.
- ✓ Correlate the strength of an acid to the strength of its conjugate base.
- ✓ Explain how the equilibrium position of a proton-transfer reaction relates to the strengths of the acids and bases involved.
- ✓ Describe the autoionization of water and explain how $[H_3O^+]$ and $[OH^-]$ are related via K_w .
- ✓ Calculate the pH of a solution given $[H_3O^+]$ or $[OH^-]$.
- ✓ Calculate the pH of a strong acid or strong base given its concentration.
- ✓ Calculate K_a or K_b for a weak acid or weak base given its concentration and the pH of the solution, and vice versa.
- ✓ Calculate the pH of a weak acid or weak base or its percent ionization given its concentration and K_a or K_b . Calculate K_b for a weak base given K_a of its conjugate acid, and similarly calculate K_a from K_b .
- ✓ Predict whether an aqueous solution of a salt will be acidic, basic, or neutral.
- ✓ Predict the relative strength of a series of acids from their molecular structures.
- ✓ Define and identify Lewis acids and bases.

Key Equations

- $K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = [\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$ [16.16] Ion-product constant of water at 25 °C
- $\text{pH} = -\log[\text{H}^+]$ [16.17] Definition of pH
- $\text{pOH} = -\log[\text{OH}^-]$ [16.18] Definition of pOH
- $\text{pH} + \text{pOH} = 14.00$ [16.20] Relationship between pH and pOH
- $K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$ or $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$ [16.25] Acid-dissociation constant for a weak acid, HA
- $\text{Percent ionization} = \frac{[\text{H}^+]_{\text{equilibrium}}}{[\text{HA}]_{\text{initial}}} \times 100\%$ [16.27] Percent ionization of a weak acid
- $K_b = \frac{[\text{BH}^+][\text{OH}^-]}{[\text{B}]}$ [16.34] Base-dissociation constant for a weak base, B
- $K_a \times K_b = K_w$ [16.40] Relationship between acid- and base-dissociation constants of a conjugate acid-base pair
- $\text{p}K_a = -\log K_a$ and $\text{p}K_b = -\log K_b$ [16.41] Definitions of $\text{p}K_a$ and $\text{p}K_b$